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Biodiversità della meiofauna in praterie di fanerogame marine del Mar Adriatico Centrale (Gabicce Mare)

Meiofauna biodiversity in seagrass meadows of the Central Adriatic Sea (Gabicce Mare)

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PREFACE

This thesis was conducted as part of the European project **DiverSEA** (Integrated Observation, Mapping, Monitoring and Prediction for Functional Bio**DIVER**sity of Coastal **SEAs**), whose main purpose is to create an integrated architecture for the observation, monitoring, mapping and prediction (OMoMaP) of functional biodiversity in European coastal areas, combining emerging techniques (e-DNA/e-RNA, DNA marks, robotics, sensors, remote sensing, etc.) and traditional approaches (classical taxonomy using microscopy, molecular analyses, etc.), also using data platforms, Artificial Intelligence (AI) and Machine Learning (ML) tools. By combining all these different techniques, DiverSEA aims to bridge the various knowledge gaps associated with the single techniques.

The Polytechnic University of Marche (UNIVPM) is the only Italian research partner in the DiverSEA project. Its work consists of mapping habitats covered by the Habitats Directive 92/43/EEC, such as macroalgal forests, mussel beds, vegetated sandbanks with seagrass meadows, and their associated biodiversity. To do this UNIVPM conducted field work activities by snorkelling and scuba diving surveys, supported by UAV imaging techniques (drones) and ROV surveys, collecting sediment samples for the analyses of the associated biodiversity using traditional approaches as microscopy, molecular analyses, etc., including e-DNA/e-RNA to obtain an integrated and complete mapping of these habitats and their associated microbiomes, flora and fauna. All this information will better define their environmental status in accordance with MSFD 2008/56/EC and facilitate policymakers.

This thesis will analyse meiofaunal biodiversity in a coastal area of the Central Adriatic Sea (Gabicce Mare) characterised by the presence of seagrass meadows (*Zostera marina*, *Zostera noltii*, *Cymodocea nodosa*), mainly using traditional taxonomy through microscopy

along with the investigation of the environmental conditions based on the grain size and biochemical composition of organic matter in the sediment, and the habitat mapping conducted by drones, ROVs and photo-quadrats.

ABSTRACT

Seagrass meadows are key coastal ecosystems that enhance benthic biodiversity and regulate fundamental ecological processes. Meiofauna play a central role within these habitats, contributing to organic matter processing and energy transfer to higher trophic levels. This study investigated biodiversity patterns, focusing on how habitat structure, sediment characteristics, and trophic variables influence meiofaunal abundance, biomass, taxonomic richness, and assemblages' composition. Sediment samples were collected in a shallow costal area of the Central Adriatic Sea (Gabicce Mare), along five coast-to-offshore transects, both in vegetated and unvegetated sediments. Meiofauna were analysed together with sediment grain size and organic matter quantity and quality indicators, using univariate and multivariate statistical analyses to assess significant spatial patterns and possible environmental drivers. Vegetated sediments showed a more heterogeneous grain-size distribution, with a higher contribution of fine particles, and higher organic matter concentrations, as indicated by biopolymeric carbon and phytopigments distributions. Meiofaunal abundance, biomass and diversity were consistently higher in presence of seagrasses and assemblages' composition differed between vegetated and unvegetated sediments. *Nematoda*, and secondarily *copepoda*, dominated all assemblages, reflecting their ecological versatility and key role within benthic food webs. Seagrasses presence or absence was identified as the main factor explaining meiofaunal variability, while sediment grain size and trophic availability were shown to be secondary drivers, also being conditioned by the presence or absence of seagrasses. This study highlights the pivotal role of seagrass meadows in sustaining benthic biodiversity and ecosystem functioning, emphasizing the ecological consequences of seagrass loss and the need to prioritize their conservation and restoration.

RIASSUNTO

Le praterie di fanerogame marine sono ecosistemi costieri fondamentali, capaci di aumentare la biodiversità bentonica e regolare processi ecologici chiave. La meiofauna svolge un ruolo centrale all'interno di questi habitat, contribuendo al processamento della materia organica e al trasferimento di energia ai livelli trofici superiori. Questo studio ha indagato i pattern di biodiversità delle comunità di meiofauna, valutando come driver quali la struttura dell'habitat, caratteristiche dei sedimenti e fattori trofici influenzino abbondanza, biomassa, ricchezza tassonomica e composizione delle comunità. I campioni di sedimento sono stati raccolti in un'area costiera dell'Adriatico centrale (Gabicce Mare), lungo cinque transetti costa-largo, su sedimenti sia vegetati che non vegetati. Abbondanza, biomassa e ricchezza tassonomica della meiofauna sono state analizzate insieme alla granulometria e agli indicatori della quantità e qualità della materia organica, applicando analisi statistiche univariate e multivariate, per identificare i pattern spaziali significativi e i principali driver ambientali. I sedimenti vegetati hanno mostrato una distribuzione granulometrica più eterogenea, con un maggiore contributo di particelle fini e concentrazioni più elevate di materia organica, in termini sia di carbonio biopolimerico che di fitopigmenti. L'abbondanza, la biomassa e la diversità della meiofauna sono risultate più elevate in presenza di fanerogame marine e la composizione della comunità è apparsa diversa tra sedimenti vegetati e non vegetati. *Nematoda*, e secondariamente *copepoda*, sono risultati i *taxa* dominanti in tutti gli assemblaggi, riflettendo così la loro versatilità ecologica e il ruolo chiave all'interno delle reti trofiche bentoniche. La presenza o assenza delle fanerogame marine è stata identificata come il principale fattore in grado di spiegare la variabilità della meiofauna, mentre la granulometria e la disponibilità trofica sono risultate driver secondari, anch'essi condizionati dalla presenza o assenza delle fanerogame. Questo studio evidenzia il ruolo cruciale delle praterie di fanerogame marine nel sostenere la biodiversità bentonica e il funzionamento degli ecosistemi, sottolineando le conseguenze ecologiche della loro perdita e la necessità di dare priorità alla loro conservazione e al loro ripristino.

1. INTRODUCTION

1.1 Seagrass meadows and their ecological importance

Seagrasses are unique marine flowering plants, living fully submerged in shallow coastal areas (Dennison, 2009). Seagrasses are monocot angiosperms occurring worldwide, except Antarctica (Den Hartog & Phillips, 2001), generally occupying 0-30 m littoral fringes although some species can reach 50-60 m depth. Each plant consists of a rhizome with shoots bearing leaves and roots, forming repeated units, called ramets or modules, giving rise to a “modular architecture”. Rhizomes divide in branches, and each branched-rhizome system can hold many genetically identical shoots, interconnected as a single individual (Terrados & Borum, 2004). Seagrasses usually grow on soft substrates but there can be exceptions, as *Posidonia oceanica*, which can grow on rocky substrates too. Marine phanerogams are also considered qualifying elements of the environmental state (Directive 92/43 EU). Their reproduction occurs both sexually, through pollination and production of flowers and fruits, and vegetatively, through rhizome growth. Seagrasses exhibit high diversity, originating from three distinct lineages (*Cymodiaceaceae*, *Hydrocharitaceae* and *Zosteraceae*) arising from terrestrial ancestors reinvading the marine environment, leading to divergent growth strategies (Dennison, 2009). Of the ~60 species worldwide, four are native of European waters: *Zostera marina*, *Zostera noltii*, *Cymodocea nodosa* and *Posidonia oceanica*, all present in the Mediterranean Sea (Borum & Greve, 2004).

- *Zostera marina* forms dense beds from the intertidal zone up to 10-15 m depth in clear water, growing slowly and preferring fine sediments (Marba et al., 1996; Dawes & Guiry, 1992; Terrados & Borum, 2004).

- ***Zostera noltii*** is similar to *Z. marina* but has thinner leaves, occupies shallower areas (< 5 m), tolerates larger fluctuations in light availability, grows faster and has a reproductive strategy typical of pioneer and colonising species, thriving in transitional low-salinity zones (Terrados & Borum, 2004; Vermaat & Verhagen, 1996; Davison & Hughes, 1998)
- ***Cymodocea nodosa*** has a broad ecological range (from shallow subtidal up to 50-60 m), preferring high salinity and coarse sediments (Terrados & Borum, 2004).
- ***Posidonia oceanica*** is endemic of the Mediterranean Sea, forming dense and slow-growing meadows from the surface up to 40–50 m depth and is highly sensitive to anthropogenic disturbances (Terrados & Borum, 2004); it is an important bio-constructor and its meadows, because of their great importance, are listed as Priority Habitats under the Habitat Directive, Annex I, code: 1120 (EU 92/43/CEE, 1992).
- Additionally a Lessepsian-origin seagrass, ***Halophila stipulacea***, occurs in the eastern Mediterranean basin (Ruggiero & Procaccini, 2004).

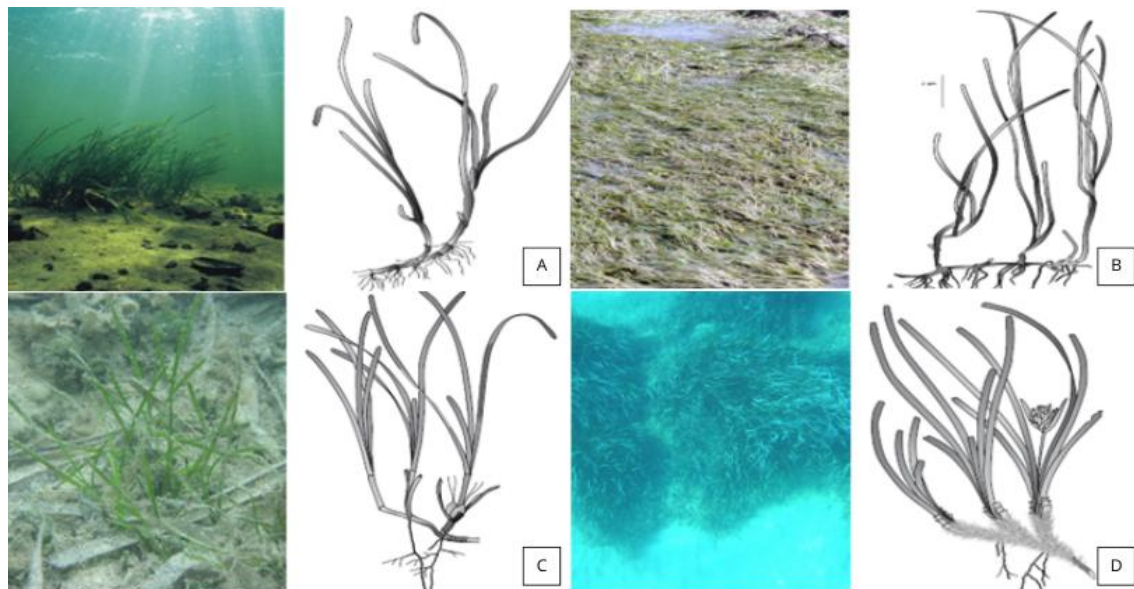


Figure 1 Seagrass species in the Mediterranean Sea. **A)** *Zostera marina* (Photo: P.B. Christensen; drawing: Dawes 1981); **B)** *Zostera noltii* (Photo: J. Borum; drawing: redrawn from NN); **C)** *Cymodocea nodosa* (Photo: O. Pedersen; drawing: redrawn from Luque and Templado 2004); and **D)** *Posidonia oceanica* (Photo: P.B. Christensen; drawing: redrawn from Luque and Templado).

Seagrasses form extensive underwater meadows, having enormous ecological importance and playing a key role in the ecosystem services and goods provisioning (Bianchelli et al., 2024). Seagrass meadows cover about 0.1- 0.2% of the global ocean (Duarte, 2002) and the estimate of their global spatial distribution, ranges from 177.000 to 600.000 km² (McKenzie et al., 2020). Seagrass growth demands an underwater irradiance generally in excess of 11% of the incident one in the water surface and this requirement typically sets their depth limits (Duarte, 1991) together with their need for a sufficient immersion into the seawater and their tolerance to waves disturbance (Hemminga & Duarte, 2000). Seagrasses can also live in estuarine and brackish waters, but they require a salinity excess of at least 5‰ for their development (Hemminga & Duarte, 2000). Seagrass habitats are highly productive (Duarte, 2002) and seagrass meadows act as bioengineers, creating complex three-dimensional habitats, supporting a high biodiversity of the associated flora, fauna and microbes (Unsworth et al., 2019), with positive implication for fisheries too (Tuya et al., 2014). The leaf canopy and the rhizomes-roots network provide substrates for attachment, stabilize the sediment, prevent resuspension and reduce irradiance, producing a microhabitats array, which are not present in unvegetated sea bottoms (Terrados & Borum, 2004). These three-dimensional habitats provide shelter and protection from predators, making seagrass meadows nurseries for several vertebrates and invertebrates, supporting complex benthopelagic food webs (Da Ros et al., 2021). Seagrass meadows are furthermore involved in coastal erosion reduction, mitigating hydrodynamic loads; seagrasses increase water quality and transparency too, provide Oxygen to both the water column and sediments, act as pollution buffers, support nutrient cycling and are very involved in carbon sequestration (Nellemann & Corcoran, 2009; Dahl et al., 2016; Green et al., 2003). Seagrass meadows are often the dominant primary producers in coastal areas, being considered as one of the most

productive ecosystem in the world (Costanza et al., 1997; Duarte & Chiscano, 1999). Below-ground biomass represents 15-50% of the seagrass net production (Duarte & Chiscano, 1999), thus limiting direct herbivory (Cebrián et al., 1997), also thanks to leaf structural compounds (such as cellulose) and phenolics, making them less palatable (Gambi et al., 2000). On the other hand, there are many organisms (especially meio- and macrofauna) feeding on epiphytes and rhizomes, while plant residues sustain the detritus chain, thus supporting high microbial biomasses, conveying organic matter and energy to higher trophic levels (Moore et al., 2004). Moreover, seagrasses contribute substantially to total ocean carbon storage (~15%), with slow decomposition rates and significant export to adjacent ecosystems (Harrison, 1989; Enríquez et al., 1993; Cebrián et al., 1997; Duarte & Chiscano, 1999). Seagrasses-dominated habitats provide essential functions and services to ocean ecosystems and human well-being (Duarte et al., 2008; Boström et al., 2011) with the global seagrass meadows natural capital which has been estimated to be ~US \$ 19.002 ha⁻¹ y⁻¹ (Costanza et al., 1997).

However, seagrass systems are extremely vulnerable to multiple direct and indirect impacts, both of natural and anthropogenic origin, also due to their distribution, which is mainly along coastal areas (Unsworth et al., 2014). Indeed, it has been estimated a seagrass loss of ~30% since 1900 (de Los Santos et al., 2019), placing this fragile ecosystem among the most threatened in the world (de Los Santos et al., 2019; Waycott et al., 2009).

Seagrass meadows can suffer for mechanical damages, coastal construction, pollution, eutrophication, fish farming and introduction of non-indigenous species (Boudouresque et al., 2009; Duarte, 2002; Duarte et al., 2020; Nakaoka & Aioi, 1999; Orth et al., 2006);

sporadic natural events can damage them too and climate change can amplify and accelerate their decline (Preen et al., 1995; Marba & Duarte, 1995; Short & Neckles, 1999).

The global seagrass meadows loss reached unprecedented rates, with IUCN indicating that 31% of seagrass species are in global decline, while 22% lack for a proper assessment (Rock & Daru, 2021), leading to declines in biodiversity, sediment protection and coastal ecosystem services, promoting shifts in primary producer dominance that cannot compensate for the ecological role of seagrasses (Borum, 1996; Burgos et al., 2017).

1.2 Meiofaunal assemblages

“*Meiobenthos*” refers to benthic organisms of intermediate size between micro- and macrobenthos (20-30 μm to 500-1000 μm ; Mare, 1942), while “meiofauna” designates only its animal component. Functionally meiofauna comprises small benthic metazoans (0.01–0.5 μg biomass), distinguished by their evolutionary history and trophic traits from other animal groups (Warwick & Clarke, 1984). Meiofauna is ubiquitous and distributed on global scale, despite the presence of significant fluctuations in its organism density. Indeed, it occurs in various environments, including marine and freshwater habitats, coastal and abyssal zones, and different sediment types (Danovaro et al., 2003). Meiofauna organisms live in close contact with sediments and exhibit functional and morphological adaptations related to substrate characteristics, they represent the most abundant group of metazoans composing benthos, with densities ranging from 100 to 1000 individuals/10 cm^2 and biomass up to 1-2 g/m^2 in coastal areas (Coull & Bell, 1979). However, these values fluctuate with latitude, sediment type, depth, hydrodynamics, salinity, temperature, nutrient availability and seasonality (Coull, 1988). Two functional categories can be distinguished: Temporary meiofauna, being within the *meiobenthos* size range only during juvenile stages, before

entering macro- or megafauna, and Permanent meiofauna, which remain within the *meiobenthos* size range throughout their whole life cycle (Danovaro & others, 2013). Meiofauna is distributed both horizontally and vertically within sediments; horizontally, its abundance is influenced by food availability, macrofauna presence, and spatial heterogeneity (Coull, 1988) whereas vertically organisms are concentrated in the top 2 cm of sediment, decreasing exponentially with depth (Shirayama, 1984), often controlled by the Redox Potential Discontinuity (RPD) layer, representing the boundary between the oxygenated and anoxic portion of the sediment (Vincx, 1996). In the first 2 centimetres we generally find oxygenation conditions with a redox potential higher than +400 mV and, when this potential falls below +200 mV, organisms density rapidly decreases (Danovaro et al., 2002). Some meiofauna organisms seem to be adapted to extreme conditions, tolerating hypoxic and even anoxic conditions, by penetrating below the reduced layer. These organisms, adapted to live in anoxic environments, with sulfuric acid production, belong to the so-called “*Thiobios*” (Krieger et al., 2000; Wetzel et al., 1995, 2002; Windoffer et al., 1999). Meiofauna displays high species and functional diversity, short generation times, limited dispersal, and high temporal variability, conferring both sensitivity and resilience to environmental changes (Gerlach, 1971; McIntyre, 1969; Herman & Heip, 1988). Ecologically, meiofauna plays a central role in benthic food webs, conveying organic matter to higher trophic level consumers and making detritus available again to macro-consumers. It also represents the linkage between the microbial loop and higher trophic levels (Coull, 1988), being the main energy-transfer pathway from the microbial world to the upper trophic levels (Danovaro, 1996; Danovaro et al., 2007, 2023).

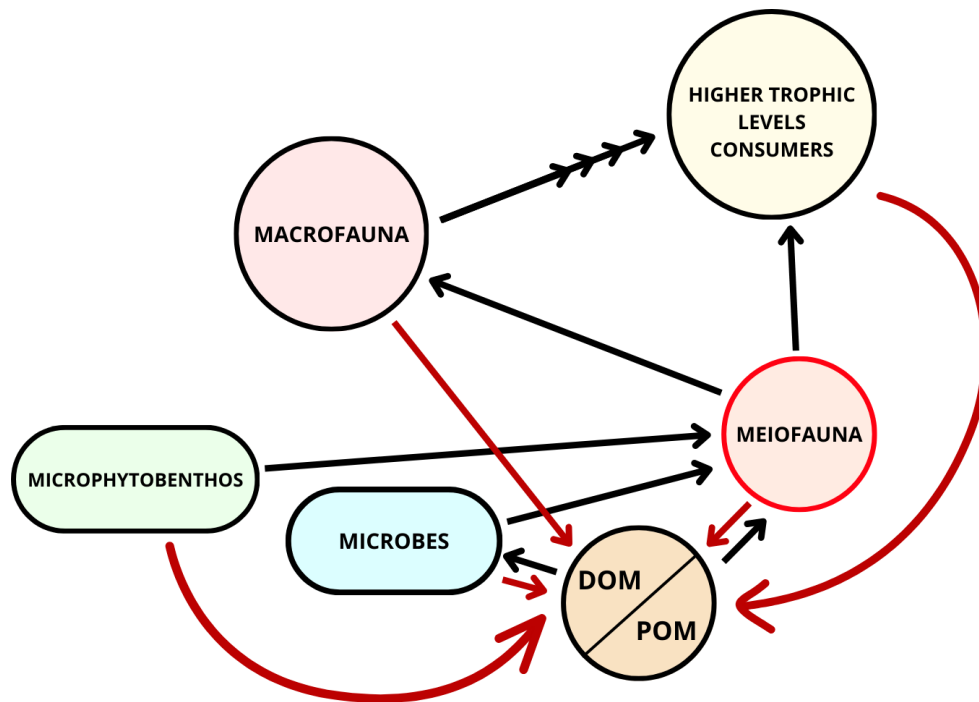


Figure 2 *Meiofaunal role within the benthic marine food web*

Of the 35 metazoan *phyla*, 22 are included in meiofaunal assemblages, with *nematoda* typically dominant (> 50%), followed by *copepoda*, *polychaeta*, and *turbellaria* (Coull, 1988). Due to their sensitivity to sedimentary changes, meiofauna (especially *nematoda* and *copepoda*) is widely used as a biological indicator of marine ecosystem health (Coull, 1999).

Seagrass ecosystems are biodiversity reservoirs, providing many ecosystem services with meiofauna being a major component, yet often neglected, of their associated fauna, playing a fundamental role in many seafloor processes, with it study which can be useful in addressing both theoretical and applied questions in ecology, conservation and evolution (García-Gómez et al., 2022; Curini-Galletti et al., 2012, 2020; Martínez et al., 2019; Schratzberger & Ingels, 2018). Meiofauna inhabits seagrass meadows ecosystems, crawling on leaves or across the sediment grains trapped among seagrasses rhizomes and roots, they

can also slip in the interstices of the adjacent sediments or drift within the detritus over the sea bottom (Garcia-Gomez et al., 2022), with both biotic (food availability, species interactions) and abiotic factors (sediment characteristics, physical stressors) influencing the associated meiofauna distribution (Mascart et al., 2015; Novak, 1982, 1989; Pusceddu et al., 2016). Studies indicate that meiofaunal abundance and diversity are higher in or nearby seagrass meadows ecosystems, compared with other marine habitats, highlighting their ecological importance (Gerovasileiou et al., 2016; Gerovasileiou & Bianchi, 2021).

AIMS OF THE PRESENT STUDY

The aims of this thesis are to study the biodiversity spatial patterns and gradients of the meiofaunal assemblages in the Gabicce Mare area (Central Adriatic Sea), hosting seagrass meadows (*Z. marina*, *Z. noltii*, and *C. nodosa*), and to identify the possible drivers influencing these patterns in this shallow coastal area.

The various environmental factors investigated as possible drivers included the presence or absence of seagrass meadows, sediment grain size, availability of trophic resources (in terms of both quantity and quality of potential food sources), and the bathymetric gradient.

These environmental variables were then related to patterns resulting from meiofaunal descriptor analyses, to assess their potential influence.

3. MATERIALS AND METHODS

3.1 Study area

This study was conducted in the Gabicce Mare (PU) coastal area, along the north-western coastline of Marche region, Central Adriatic sea ($43^{\circ}57'96.70''\text{N}$ $12^{\circ}45'85.90''\text{E}$; **Figure 3**), within a Site of Community Importance. The area is delimited to the north by a touristic harbour and the Tavello River, and to the south by Mount San Bartolo, which is included within a Regional Natural Park established to protect its high naturalistic value (Colantoni et al., 2004; Gambi et al., 2024). Gabicce Mare is a benthic biodiversity hotspot, hosting seagrass meadows of *Zostera marina*, *Zostera noltii* and *Cymodocea nodosa* (Carugati et al., 2018). The study area is enclosed by breakwater barriers, located approximately 100 m offshore and parallel to the coastline (Carugati et al., 2018), significantly influencing local hydrodynamic conditions and sediment transport (Semprucci et al., 2010).

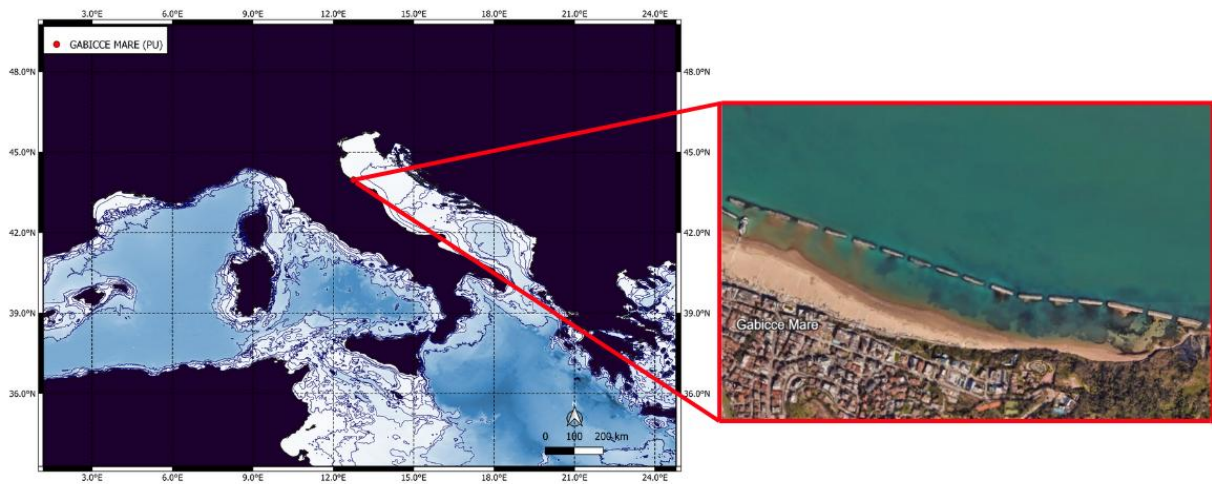


Figure 3 Study area Map, Q-GIS.

3.2 *Sampling strategy*

Sediment sampling was carried out in May 2024. Prior to sampling, the study area (15.05 hectares) was mapped using aerial drone surveys combined with underwater visual and photographic quadrat censuses, performed by scientific divers (**Figures 4 and 5**). This preliminary mapping aimed to assess the distribution and spatial extent of seagrass meadows to design the sampling strategy, according to seagrass coverage. Five coast-to-offshore transects (100 m in length) were subsequently established within the breakwater barriers, covering the entire study area from east to west. Three sampling stations were selected along each transect, resulting in a total of fifteen stations. Transects were labelled from A to E, and stations were numbered from 1 to 15, progressing westward (**Figure 6**). Depth among stations was very similar, ranging from 1 to 3.1 meters. The sampling design was structured to include both vegetated and unvegetated sediments. All stations along transect B and C were located within seagrass meadows, together with stations 2 and 3 of transect A and station 12 of transect D. In contrast, station 1 of transect A, stations 10 and 11 of transect D, and all the stations of transect E were located outside the meadow boundaries.

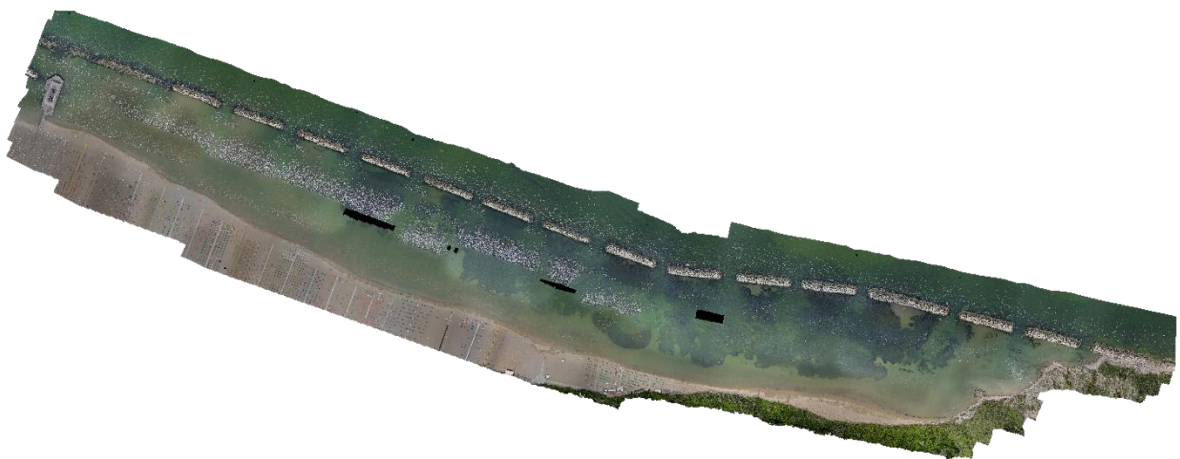


Figure 4 Orthomosaic of the study area obtained by using the drone

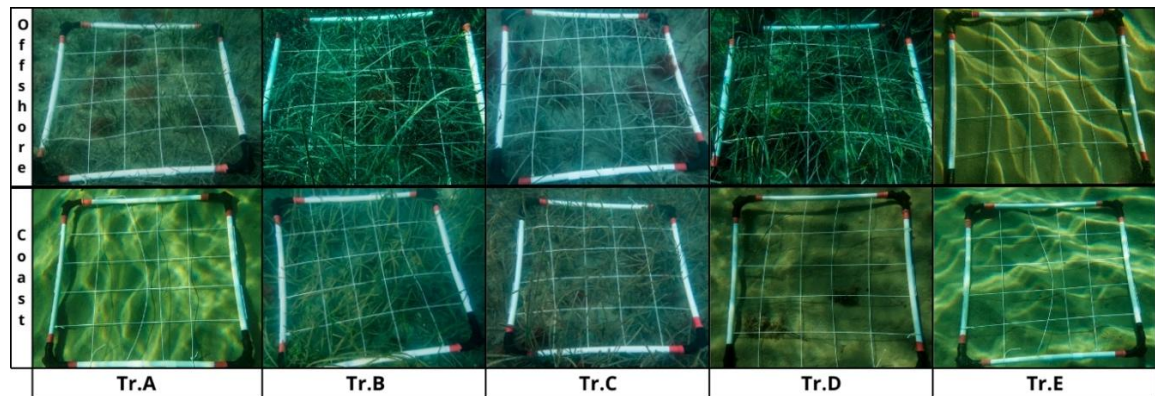


Figure 5 Examples of photo-quadrats taken along the transects, showing how the type of substrate and vegetation coverage changes between different transects and along the transects themselves.



Figure 6 Transects and sampling stations' geographical location

Sediment samples were collected using hand corers (10 cm internal diameter). For each station, three independent replicates (R1, R2, R3) were taken for each type of analysis (grain size, organic matter and meiofauna analyses). Immediately after collection, each core was vertically sectioned into three sediment layers (0–1 cm, 1–2 cm and 2–10 cm), separately

stored in pre-labelled containers and preserved at -20°C until laboratory analyses (Figure 7).



Figure 7 Field sampling activities A) Sediment core used during sampling; B) Sediment core subdivision

3.3 Laboratory analyses

3.3.1 Grain size

For the grain size analyses, it was used the Udden–Wentworth scale (Udden, 1914; Wentworth, 1922, 1934). The sediment sample was weighed and washed in H_2O_2 to remove the organic fraction and then dried in an oven (60°C for 24 hours). The arenitic fraction ($> 62.5\ \mu\text{m}$) was separated from the silt-clay fraction ($< 62.5\ \mu\text{m}$) using a specific sieve equipped with a $62.5\ \mu\text{m}$ mesh. Once dried again in an oven, the sandy fraction was separated into its components using a series of sieves, stacked in a pile, with progressively decreasing mesh sizes. Each fraction obtained was then weighed, related as a percentage to the initial total weight and referred to its respective grain size class of the Wentworth scale.

3.3.2 Biochemical composition of sedimentary organic matter

3.3.2.1 Phytopigments

Chlorophyll-a and phaeopigments were extracted using the method described by Lorentzen and Jeffrey (Lorenzen & Jeffrey, 1980). Pigments were extracted (12 hours at 4°C in the dark) from sediment samples, using 5 ml of 90% acetone. After extraction, a fluorimetric analysis was performed to estimate the concentration of chlorophyll-a ($\mu\text{g/g}$), followed by acidification by adding 200 μl 0.1 M HCl to estimate the concentration of phaeopigments ($\mu\text{g/g}$). The total phytopigments concentrations ($\mu\text{g/g}$) were obtained by adding chlorophyll-a and phaeopigments ones, by using the conversion factor of 40 μg of C/ μg (De Jonge, 1980) to obtain the carbon equivalent.

3.3.2.2 Total Proteins

Total protein was determined using the method proposed by Hartree (1972), which is a modification of the procedure proposed by Lowry et al. (1951), adapted by Danovaro & Fabiano (1990) for the analysis of marine sediments. This procedure uses a colorimetric method producing a stable blue colouration, proportional to the protein content of the reaction mixture. The reading was taken using a spectrophotometer at a wavelength of 650 nm and to calculate the protein concentrations (mg/g), the values were plotted on a calibration curve, obtained from a standard bovine albumin solution.

3.3.2.3 Total Carbohydrates

Total carbohydrates were determined using the method developed by DuBois et al. (1956), adapted to marine sediments by Gerchakov & Hatcher (1972). This procedure is based on the carbohydrate-phenol reaction in presence of concentrated sulphuric acid. The reading

was taken using a spectrophotometer at a wavelength of 600 nm, with carbohydrate concentrations (mg/g) which were calculated using calibration curves, obtained from a standard solution of D (+) glucose.

3.3.2.4 *Total Lipids*

Total lipid determination was performed using the Bligh & Dyer (1959) method, which exploits the solubility of lipids in organic solvents such as methyl alcohol and chloroform, adapted to sediments by Danovaro & Fabiano (1990). Lipids were extracted from sediments using methanol and chloroform (2:1 vol/vol) and then reacted with concentrated sulphuric acid in a carbonisation reaction. The reading was taken using a spectrophotometer with a wavelength of 375 nm. The lipid concentration (mg/g) was calculated based on calibration curves obtained from the reaction in chloroform of standard tripalmitin solutions.

3.3.2.5 *Biopolymeric Carbon*

Biopolymeric carbon (BPC) has been defined as a measure of sedimentary organic carbon potentially available to benthic heterotrophic organisms. BPC is given by the sum of the three main categories of biochemical compounds of the organic matter (carbohydrates, lipids and proteins) converted to equivalent carbon using relative conversion factors, respectively 0.49, 0.40 and 0.75 μg of C/ μg (Fabiano & Danovaro, 1994).

3.3.2.6 *Quality indicators of organic matter*

The contribution of autotrophic carbon (from phytopigments) to total BPC, the protein-to-carbohydrate ratio and the chlorophyll-a-to-phaeopigments ratio can be used as descriptors of the bioavailability of organic matter in sediments (Pusceddu et al., 2009). The contribution of autotrophic carbon to BPC indicates the amount of carbon of primary origin that

contributes to the bioavailability of organic carbon, after converting the concentration of chlorophyll-a into carbon equivalent. Since photosynthetic pigments are labile components, their carbon content relative to BPC indicates the freshness of the organic matter (Dell'Anno et al., 2002). The protein-to-carbohydrate ratio indicates the presence of refractory organic matter (Danovaro et al., 1993), since proteins are more labile and more rapidly mobilised than carbohydrates, so we can use this ratio as an ageing index, reflecting the organic substance freshness and it is also used to classify the benthic trophic status (Dell'Anno et al., 2002). Chlorophyll-a-to-phaeopigments ratio is another freshness index of sedimentary organic matter. Chlorophyll-a is present in all photosynthetic organisms, while phaeopigments are pigment-degradation products. We can therefore understand that a higher chlorophyll-a-to-phaeopigments ratio indicates greater freshness of organic matter and vice versa (Morata & Renaud, 2008).

3.3.3 Meiofauna

3.3.3.1 Extraction

Meiofauna was extracted from the sediment by density gradient centrifugation using a Ludox HS40 solution (Heip et al., 1985). The samples were initially sonicated using a sonicator to detach the sediment particles adhering to the organisms' surface. The extraction continued with the sample being filtered first through a 500 µm mesh sieve, then collected in a beaker and re-filtered through a 20 µm mesh sieve, thus isolating only organisms within the size range of meiofauna. At this point, the remaining sediment was placed in a 50 ml Falcon tube with Ludox HS40 (density 1.31 g/cm³) and centrifuged for 10 minutes at 3000 rpm. After centrifugation, the supernatant was poured into the 20 µm mesh and then rinsed with water.

This centrifugation-filtering operation is repeated for a total of three times for each sample to achieve a 98% extraction efficiency. At the end of the three centrifugation-filtering cycles, whatever remains on the 20 μm filter is transferred to a Falcon tube and fixed with 70% EtOH, waiting to be sorted and, in our case, the samples were not stained with Rose Bengal. For each sample, the R1 replica from each station was fixed with 70% EtOH, once extracted in three different Falcon tubes, maintaining the 0-1 cm, 1-2 cm and 2-10 cm stratification, while for the R2 and R3 replicas, the three layers were combined and fixed together inside the same Falcon tube to form an integrated layer.

3.3.3.2 *Sorting*

Once the organisms have been extracted from the sediment, the sample is observed using a stereoscopic microscope at 40x magnification, so that all the meiofauna *taxa* present can be identified. To perform the count, the contents of each Falcon tube were placed inside a Delfuss *cuvette*, internally divided into 200 cells (5x5 mm), arranged in 10 rows of 20 cells each. Meiofaunal abundance was obtained by comparing the number of individuals to a known sediment surface area (10 cm^2) and using the percentage contribution of each *taxon* to the total abundance of meiofauna organisms, it was possible to obtain the taxonomic composition of the assemblages.

Some meiofauna organisms were collected from the sample during sorting to be photographed using the HIROX HRX-01 digital microscope.

Diversity as *Taxa* richness was obtained by the sum of the number of *taxa* identified in the three replicates for each station.

3.3.3.3 *Meiofaunal Biomass*

An indirect volumetric method was used to determine meiobenthic organisms' biomass, in which dry weight is extrapolated from volume measurements. This method consists of associating the shape of an animal with a geometric figure, since most meiobenthic *taxa* have a regular shape. To carry out the measurements necessary to estimate the biovolume of *nematoda*, it was necessary to prepare permanent slides using the formalin-ethanol-glycerol technique (Grise, 1969; Seinhorst, 1959; Vincx, 1996). For all meiofauna organisms, the volume (**V**) was estimated by measuring the body length (**L**) and maximum width (**W**) using a binocular microscope or optical microscope (in the case of *nematoda*).

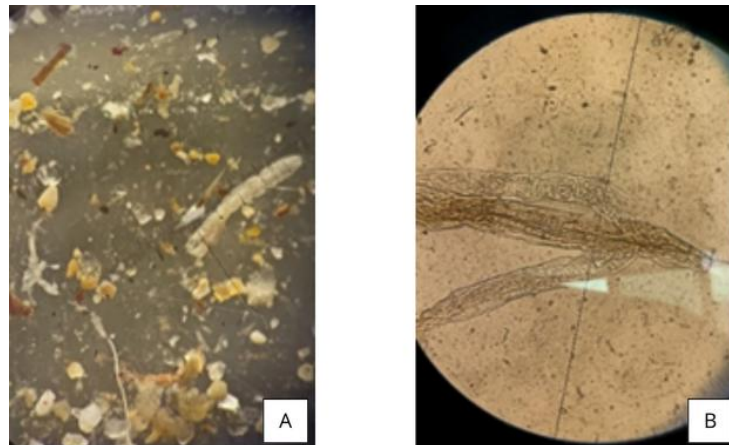


Figure 8 Examples of taxon biomass measurement: A) *Tanaidacea*; B) *Nematoda*

Once obtained individual length and width measurements, mathematical formulas were applied to calculate biovolumes. For *nematoda*, whose shape can be associated with a cylinder, the following formula, proposed by Andrassy, 1956, was applied:

$$V = L \times W^2 \times 0.063 \times 10^{-5}$$

where **V** is expressed in nl (10^{-9} litres) and **W** and **L** in μm .

For all other taxa, the formula proposed by Warwick & Price (1979) was applied:

$$V = C \times L \times W^2,$$

where **V** is expressed in nl, **L** and **W** in mm and **C** is the specific conversion factor for each *taxon* (**Table 1**).

Table 1 List of conversion of C-factors, calculated by (Warwick & Gee, 1984) to calculate the biovolume of meiobenthic organisms; each taxon has a specific C factor based on the geometric shape to which it can be approximated.

<i>Taxon</i>	<i>C</i>
<i>Nematoda</i>	530
<i>Polychaeta</i>	530
<i>Bivalvia</i>	450
<i>Ostracoda</i>	450
<i>Kinorhyncha</i>	295
<i>Oligochaeta</i>	530
<i>Tardigrada</i>	614
<i>Cumacea</i>	400
<i>Amphipoda</i>	230
<i>Isopoda</i>	230
<i>Tanaidacea</i>	400
<i>Acarina</i>	230
<i>Gasteropoda</i>	560
<i>Copepoda</i>	(*)

(*) For copepoda, the C value varies depending on the shape of the animal, as suggested by Warwick and Gee (1984), from a minimum of 230 for “discoidal” (“Scutelliform”) shapes to a maximum of 750 for “cylindrical” shapes.

Once obtained, the biovolume was converted to wet weight, assuming a specific gravity of 1.13 (Wieser, 1960). From this, we moved on to dry weight, considering it to be 25% of wet weight (Myers, 1967; Wieser, 1960). Finally, the dry weight was converted into carbon content, assuming that it is 40% of the dry weight (Feller & Warwick, 1988). The total biomass, expressed in µgC/10cm², has been obtained by adding the biomass values of *nematoda* and of the other taxa present at each station.

3.4 Statistical analyses

To assess the presence/absence of differences for all parameters among transects, PERMANOVA (Permutational Multivariate Analysis of Variance) was performed, using 999 permutations, applying a Bray–Curtis distance matrix for assemblages' composition variables, while a Euclidean distance matrix was applied to all other parameters. Significant PERMANOVA results ($p < 0.05$) were followed by pairwise tests, recalculating PERMANOVA for each pair of considered levels, corresponding to couples of transects, to determine among which levels the significant differences occur.

To determine which of the studied variables (grain size, depth, biochemical composition of sedimentary organic matter and presence/absence of seagrass beds) explained variations in meiofaunal abundance, diversity as *taxa* richness, biomass and assemblages structure of the meiofaunal assemblages along the sample, a DistLM (Distance-based Linear Model) analysis was performed with each variable tested using 999 random parameters (McArdle & Anderson, 2001).

Meiofauna assemblages were furthermore analysed using NMDS (non-metric multidimensional scaling) based on Bray–Curtis dissimilarities, applied separately to total meiofaunal abundance and biomass. NMDS is a robust, non-parametric ordination technique widely used in ecology to represent multivariate assemblages patterns and underlying ecological gradients in reduced dimensional space (Clarke, 1993; Minchin, 1987).

Ordination plots were used to visualize similarities among samples and to explore differences between vegetated and unvegetated sediments. Environmental variables were fitted inside the ordination space as vectors, to identify the main environmental factors potentially driving assemblages' patterns.

4. RESULTS

4.1 Grain size

The grain size composition is reported in **Figure 9**.

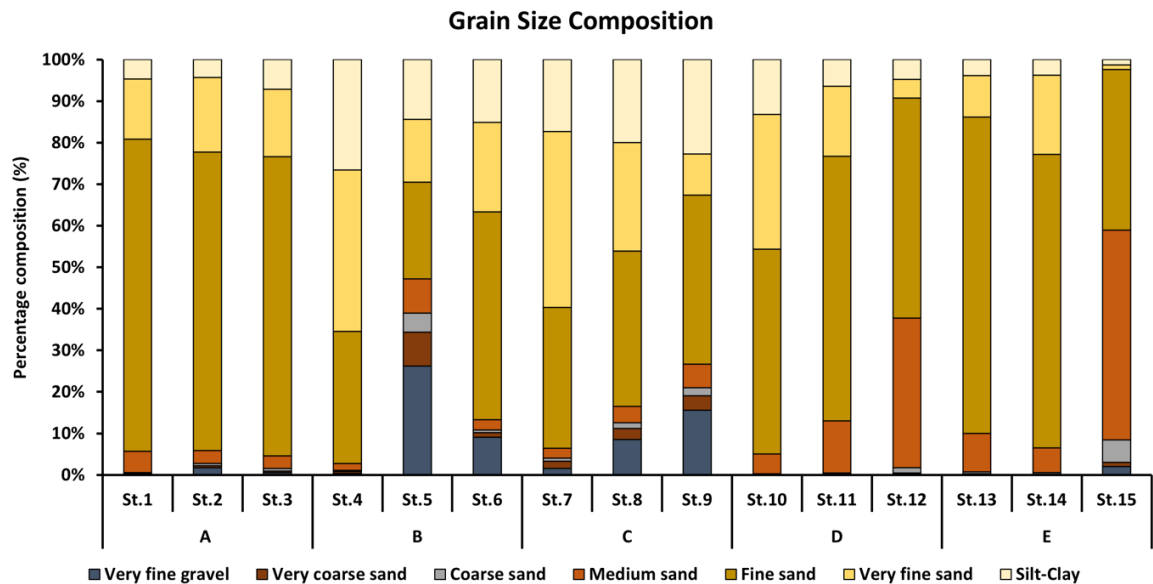


Figure 9 Grain size percentage composition

PERMANOVA revealed significant differences ($p < 0.05$) in sediment composition among transects. Transect A was characterised by a dominance of fine sand (125–250 μm ; $> 70\%$) and very fine sand (63–125 μm ; $> 15\%$), accounting together for more than 88% of the sediment. Pairwise tests indicated that transect A differed significantly from transects B and C ($p < 0.05$). Transects B and C exhibited the highest grain size heterogeneity, with relatively higher contributions of very fine gravel (2–4 mm) and silt-clay fractions ($< 63 \mu\text{m}$), although fine and very fine sands remained the dominant components. No significant differences were observed between transects B and C (pairwise tests, $p > 0.05$), whereas both differed significantly from the remaining transects. Transects D and E were characterised by higher percentages of medium sand (250–500 μm) and lower silt-clay fraction, compared with the

other transects, together with fine sand as the dominant fraction. Pairwise comparisons showed no significant differences between transects D and E ($p > 0.05$), while significant differences were observed when each was compared with all other transects ($p < 0.05$).

4.2 Biochemical composition of sedimentary organic matter

4.2.1 Phytopigments

Chlorophyll-a (**Figure 10**) and phaeopigment (**Figure 11**) concentrations exhibited comparable spatial patterns across stations.

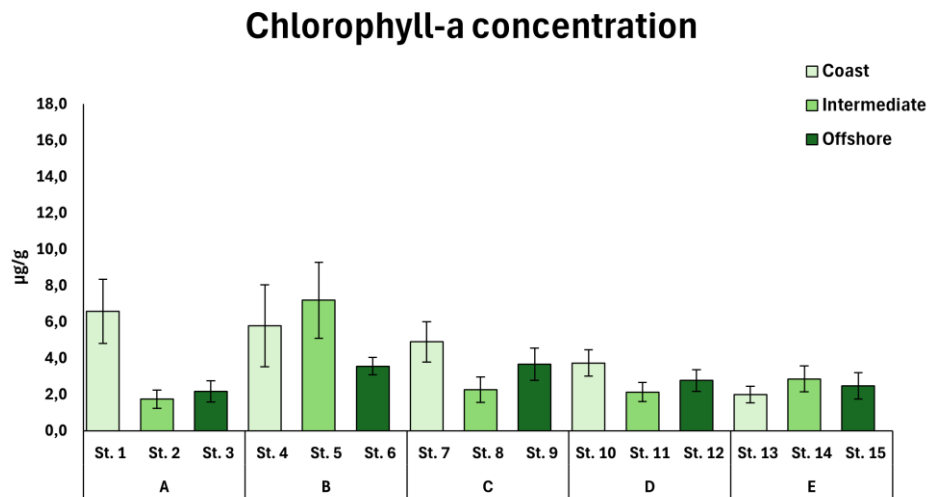


Figure 10 Chlorophyll-a concentrations (µg/g)

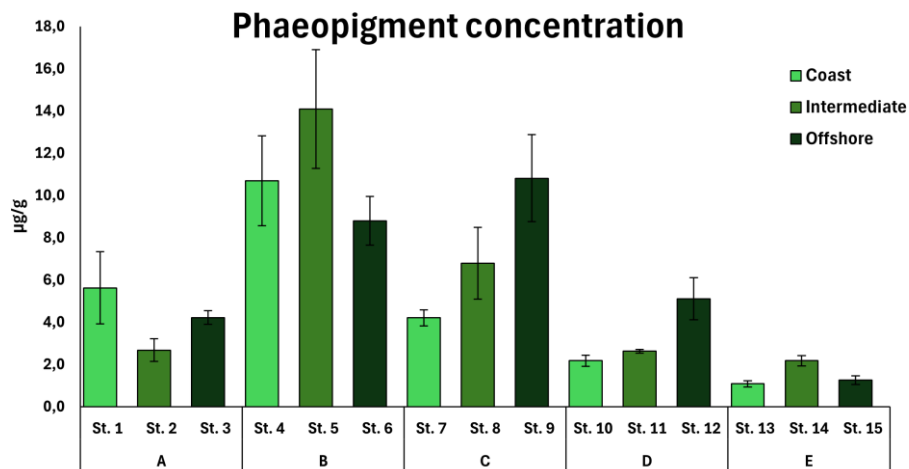


Figure 11 Phaeopigments concentration (µg/g)

Overall, phaeopigment concentrations exceeded those of chlorophyll-a, except at stations 1 and 2, where chlorophyll-a was higher. The highest concentrations of both pigments were observed in transect B, peaking at station 5 ($7.19 \pm 2.09 \mu\text{g/g}$ for chlorophyll-a and $14.09 \pm 2.81 \mu\text{g/g}$ for phaeopigments), followed by transect C. The lowest chlorophyll-a concentration was measured in transect A, station 2 ($1.74 \pm 0.50 \mu\text{g/g}$), whereas the lowest phaeopigment concentration occurred in transect E, station 13 ($1.08 \pm 0.15 \mu\text{g/g}$).

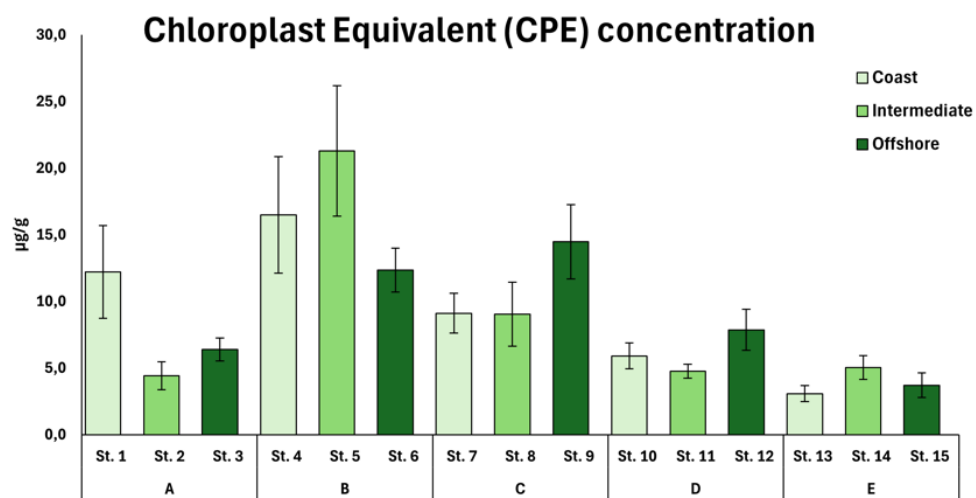


Figure 12 Total phytopigments concentration ($\mu\text{g/g}$)

The sum of chlorophyll-a and phaeopigments, expressed as total phytopigment concentration (**Figure 12**), differed significantly among transects, as indicated by PERMANOVA ($p < 0.05$). Pairwise comparisons revealed highly significant differences among transects A–B, B–D, B–E, and C–E ($p = 0.001$) while differences between transects C–D ($p = 0.006$), B–C ($p = 0.012$), and A–E and D–E ($p = 0.028$) were significant but less pronounced. No significant differences were detected between transects A–C or A–D ($p > 0.05$) and no clear coast-to-shore distribution patterns were observed along the transects.

4.2.2 Total Proteins

Total protein concentration (**Figure 13**) differed significantly among transects according to PERMANOVA ($p < 0.05$) with pairwise comparisons revealing significant differences between all transect pairs ($0.001 < p < 0.007$). The highest protein concentrations were recorded along transect B, with a maximum at station 5 (4.28 ± 0.12 mg/g), followed by transect C. In contrast, the lowest values were observed in transects D and E, with a minimum at station 15 (0.48 ± 0.02 mg/g). No clear coast-to-land distribution patterns were detected within the transects.

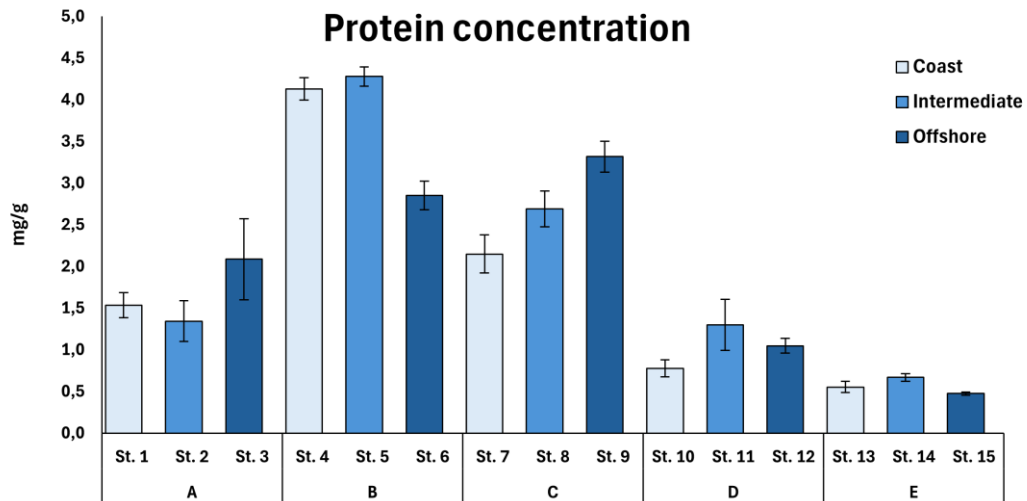


Figure 13 Total proteins concentration (mg/g)

4.2.3 Total Carbohydrates

Total carbohydrate concentration (**Figure 14**) differed significantly among transects as indicated by PERMANOVA ($p < 0.05$). Pairwise comparisons revealed significant differences among all transects ($0.001 < p < 0.004$), except for transects A and C, which did not differ significantly ($p > 0.05$), indicating similar concentrations. The highest

carbohydrate concentrations were recorded along transect B, with a maximum at station 5 (2.54 ± 0.21 mg/g), whereas the lowest values occurred in transects D and E, with a minimum at station 15 (0.22 ± 0.05 mg/g). No broad coast-to-land distribution patterns were evident along the transects.

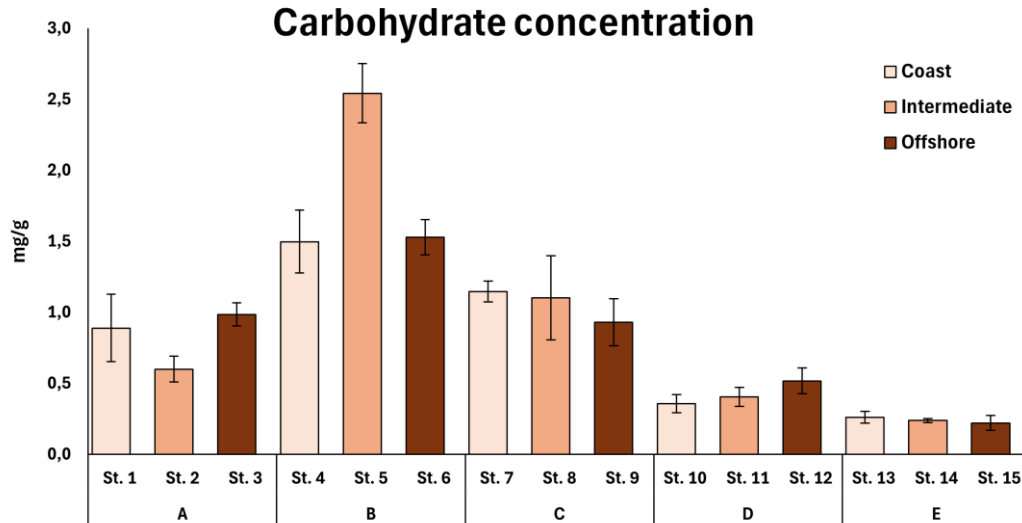


Figure 14 Total carbohydrates concentration (mg/g)

4.2.4 Total Lipids

Total lipid concentration (**Figure 15**) differed significantly among transects according to PERMANOVA ($p < 0.05$). Pairwise comparisons revealed significant differences among all transects ($p < 0.05$), except for transects A and C ($p > 0.05$). Highly significant differences were observed between the pairs A–B, A–D, A–E, B–D, B–E, C–D, and C–E ($0.001 < p < 0.004$), whereas differences between transects B–C ($p = 0.014$) and D–E ($p = 0.017$) were significant but less pronounced. Indeed, the highest lipid concentrations were recorded in transect B, peaking at station 5 (0.66 ± 0.08 mg/g), followed by transect C, while the lowest

ones were observed in transects D and E, with a minimum at station 15 (0.14 ± 0.04 mg/g).

No broad coast-to-land distribution patterns were evident for lipids along the transects.

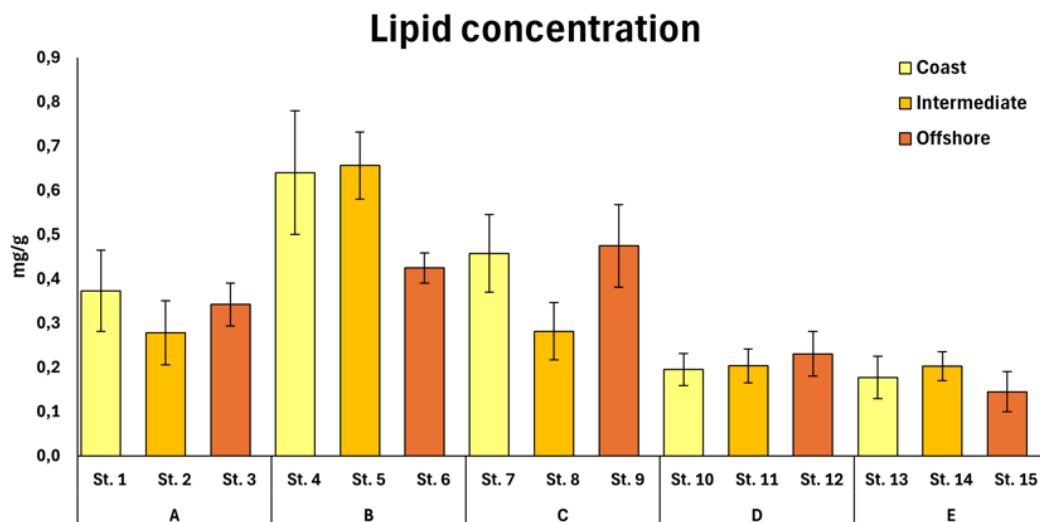


Figure 15 Total lipids concentration (mg/g)

4.2.5 Biopolymeric Carbon

Biopolymeric carbon (BPC) concentrations along the transects (**Figure 16**) differed significantly according to PERMANOVA ($p < 0.05$). Pairwise comparisons revealed significant differences between all transect pairs ($0.001 < p < 0.006$). BPC distribution mirrored that of the constituent macromolecules, with the highest concentrations observed along transect B, peaking at station 5 (3.60 ± 0.15 mg/g), followed by transect C, and the lowest values were recorded in transects D and, particularly, E, with a minimum at station 15 (0.43 ± 0.05 mg/g). As observed for proteins, carbohydrates, and lipids, no coast-to-offshore distribution pattern was evident for BPC along the transects.

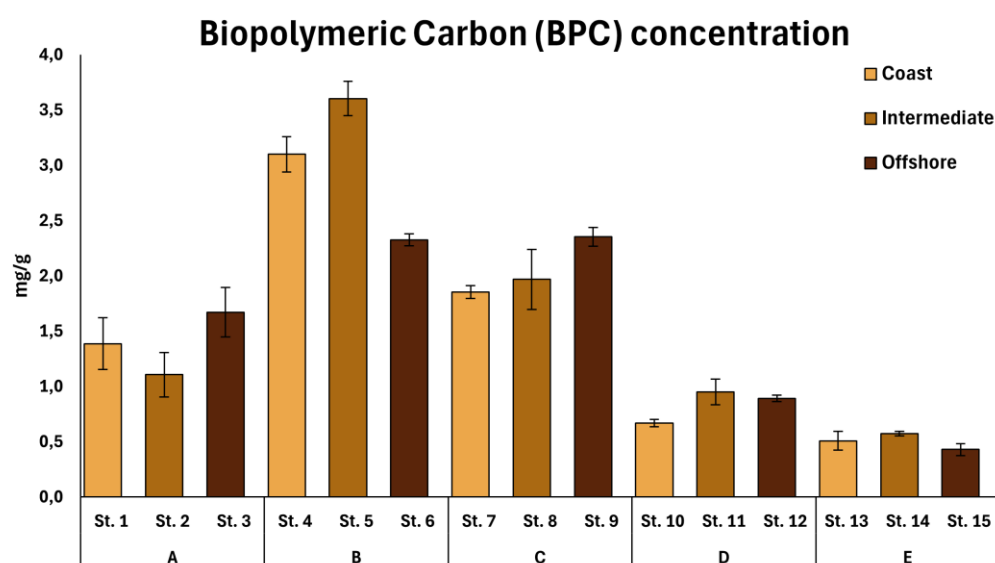


Figure 16 Total biopolymeric carbon concentration (mg/g)

4.2.6 Biochemical Composition of organic matter

The percentage of the biochemical composition of BPC across the 15 analysed stations, comparing the relative contributions of the three main biochemical components (proteins, carbohydrates, and lipids) in each station is reported in **Figure 17**.

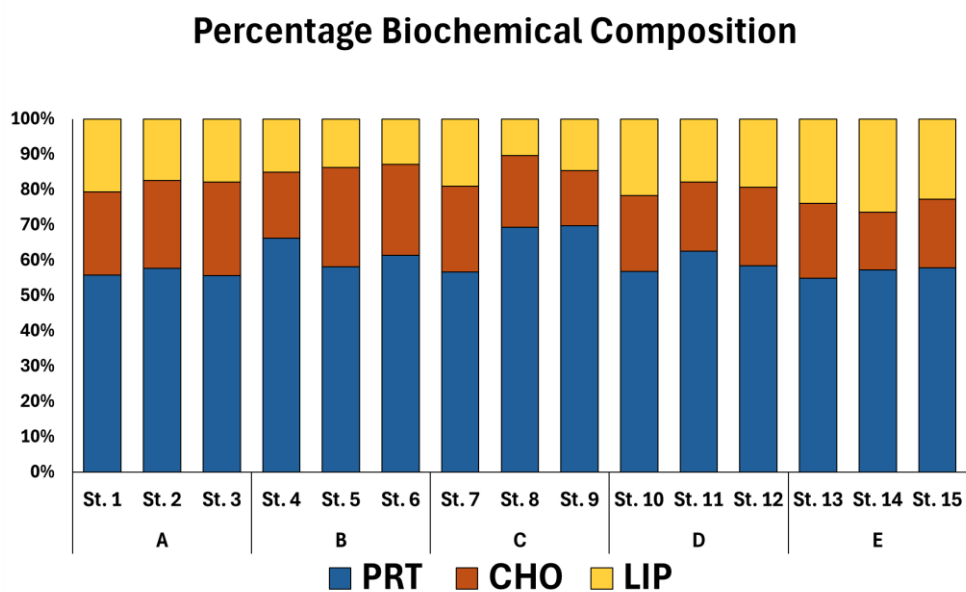


Figure 17 Biopolymeric Carbon Percentage Biochemical Composition

Proteins represented the largest fraction at all stations, exceeding 50%, with slightly higher proportions in transects B and C. In transect A, carbohydrates and lipids contributed similarly, with carbohydrates marginally more abundant. In transects B and C, carbohydrates were more prominent than lipids, whereas in transect D, lipids and carbohydrates were present in similar proportions. In transect E, lipids contributed more than carbohydrates.

4.2.7 *Quality Indicators of food sources*

Figure 18 shows the contribution of autotrophic carbon to total BPC, expressed as a percentage of primary carbon.

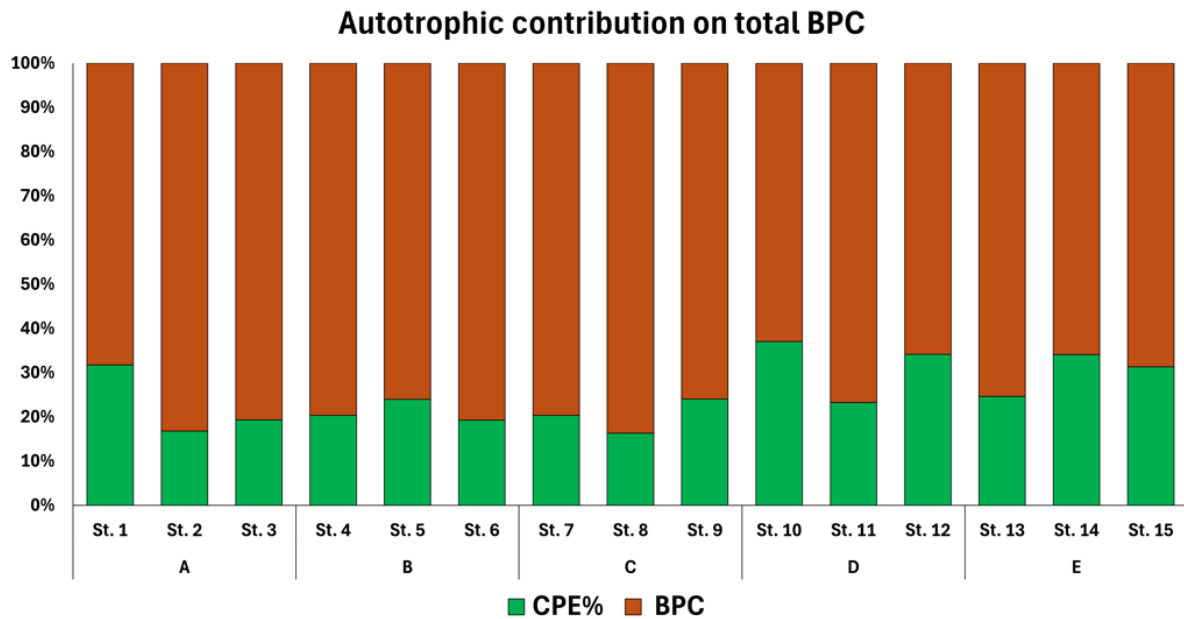


Figure 18 Autotrophic contribution on total BPC

PERMANOVA indicated significant differences among transects ($p < 0.05$), and pairwise comparisons revealed significant differences between transects B–D, B–E, C–D, C–E, and A–D. Differences among the first four pairs were more pronounced ($0.006 < p < 0.009$) than that between transects A and D ($p = 0.046$), while no significant differences were detected

among the remaining transects ($p > 0.05$), showing comparable values of primary carbon. Autotrophic carbon accounted for at least 16% at all stations, with the lowest values observed in transects B and C, reaching a minimum at station 8 (16.4%), and higher percentages in transects D and E, peaking at station 10 (37.1%).

For the protein-to-carbohydrate ratio, PERMANOVA detected no significant differences ($p > 0.05$), indicating comparable values among transects. Nevertheless, inspection of **Figure 19** shows the highest ratio in transect D at station 11 (4.91 ± 1.73 mg/g) and the lowest in transect B at station 5 (1.85 ± 0.10 mg/g). We can therefore see that in all stations we have a positive ratio due to a greater presence of proteins.

Chlorophyll-a-to-phaeopigment ratio analysis (**Figure 20**) revealed significant differences among transects, according to PERMANOVA ($p < 0.05$). Pairwise comparisons indicated highly significant differences between transects A–E, B–D, B–E, and C–E ($0.001 < p < 0.005$), as well as significant but less pronounced differences between transects A–B ($p = 0.022$), D–E ($p = 0.041$), and C–D ($p = 0.046$) with the highest ratios recorded in transect E, peaking at station 13 (1.75 ± 0.29 $\mu\text{g/g}$), followed by transect D, whereas the lowest ratios occurred in transects B and C, with a minimum at station 8 (0.28 ± 0.03 $\mu\text{g/g}$). No significant differences were detected among the remaining transects ($p > 0.05$).

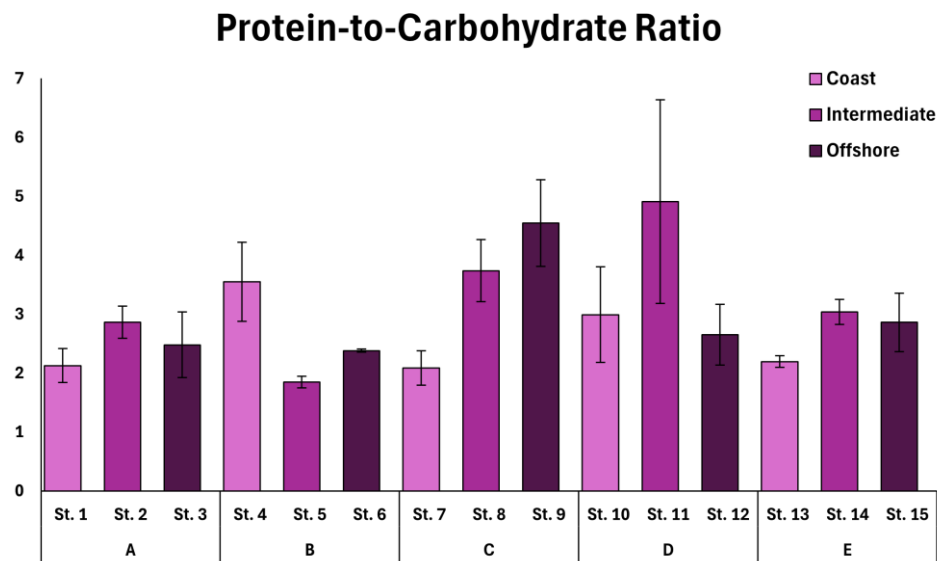


Figure 19 Protein-to-Carbohydrate ratio

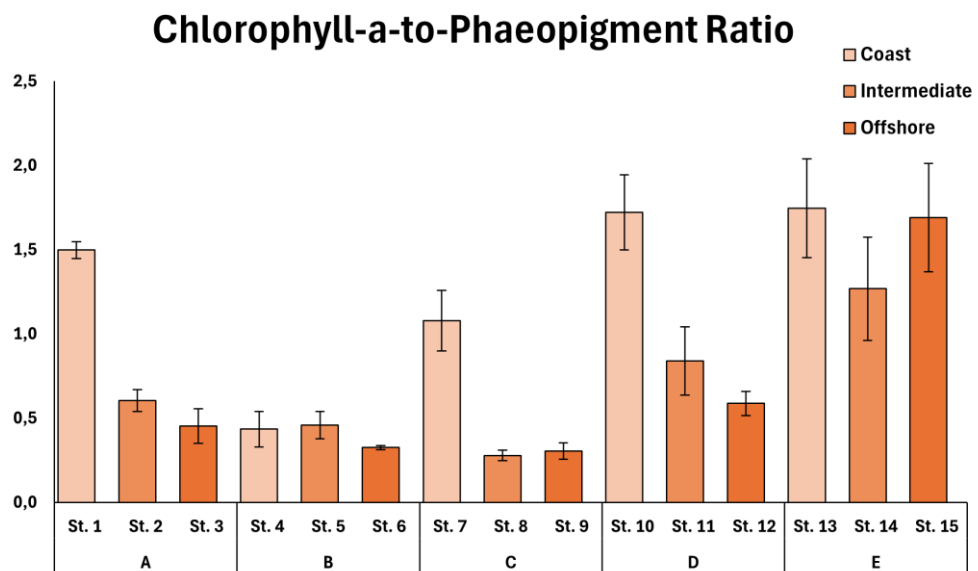


Figure 20 Chlorophyll-a-to-Phaeopigments ratio

4.3 Meiofauna

4.3.1 Meiofaunal total abundance

PERMANOVA analysis of total meiofaunal abundance revealed significant differences among transects ($p < 0.05$). Pairwise comparisons indicated highly significant differences

between transects A–D, A–E, B–D, B–E, C–D, and C–E ($0.001 < p < 0.002$), and significant but less pronounced differences between transects B and C ($p = 0.013$).

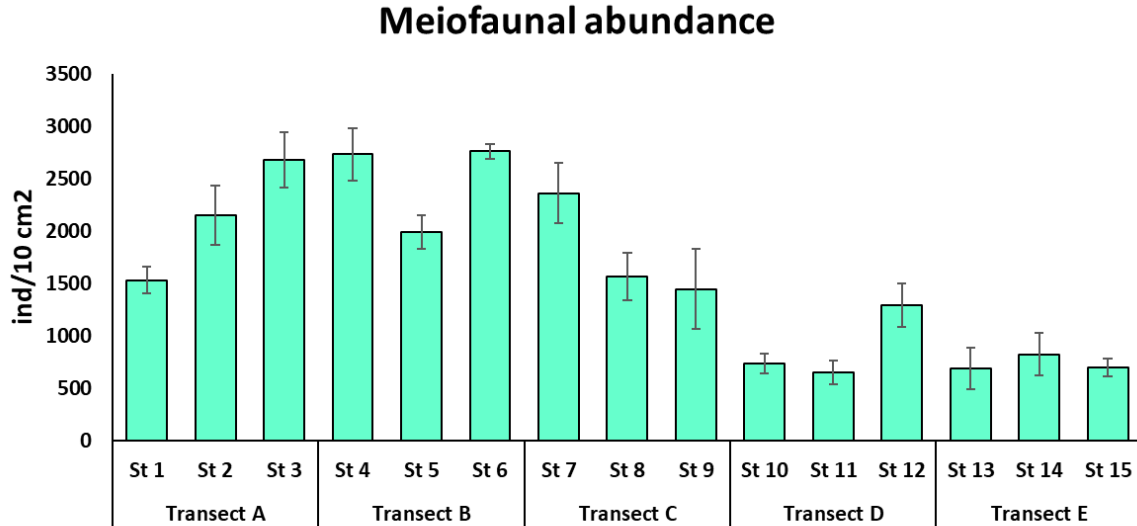


Figure 21 Meiofaunal total abundance (ind/10 cm²)

As shown in **Figure 21**, the highest total meiofaunal abundances were recorded along transect B, peaking at stations 6 (2761.3 ± 73.8 ind/10 cm²) and 4 (2732.9 ± 248.0 ind/10 cm²), while station 5 of the same transect exhibited lower values (1990.7 ± 162.1 ind/10 cm²). Transect B was followed by transects A and C in terms of total abundance, not showing significant differences, together with transect A and B ($p > 0.05$). The lowest values were observed in transects D and E, too lacking significant differences ($p > 0.05$), with a minimum at station 11 (650.7 ± 115.4 ind/10 cm²). Within these latter transects, station 12 is an exception, displaying a local increase in total meiofaunal abundance (1293.6 ± 210.3 ind/10 cm²).

4.3.2 Richness of Higher Meiofaunal Taxa

As shown in **Figure 22**, diversity as *taxa* richness did not differ significantly among transects A, B, and C ($p > 0.05$), ranging between 8 and 9 *taxa*, with local deviations observed at

station 9 (10 *taxa*), station 3 (11 *taxa*), and station 6 (12 *taxa*). The lowest *taxa* richness was recorded in transects D and E, without significant differences between them ($p > 0.05$), ranging from 4 to 6 *taxa*, except for station 12, which exhibited a local peak of 8 *taxa*. PERMANOVA analysis of *taxa* richness revealed significant differences among transects ($p < 0.05$). Pairwise comparisons showed highly significant differences between transects A–D, A–E, B–D, and B–E ($0.001 < p < 0.002$), as well as significant but less pronounced differences between transects C–E ($p = 0.009$) and C–D ($p = 0.015$). A total of 13 different meiofaunal *taxa* were recorded across the study area.

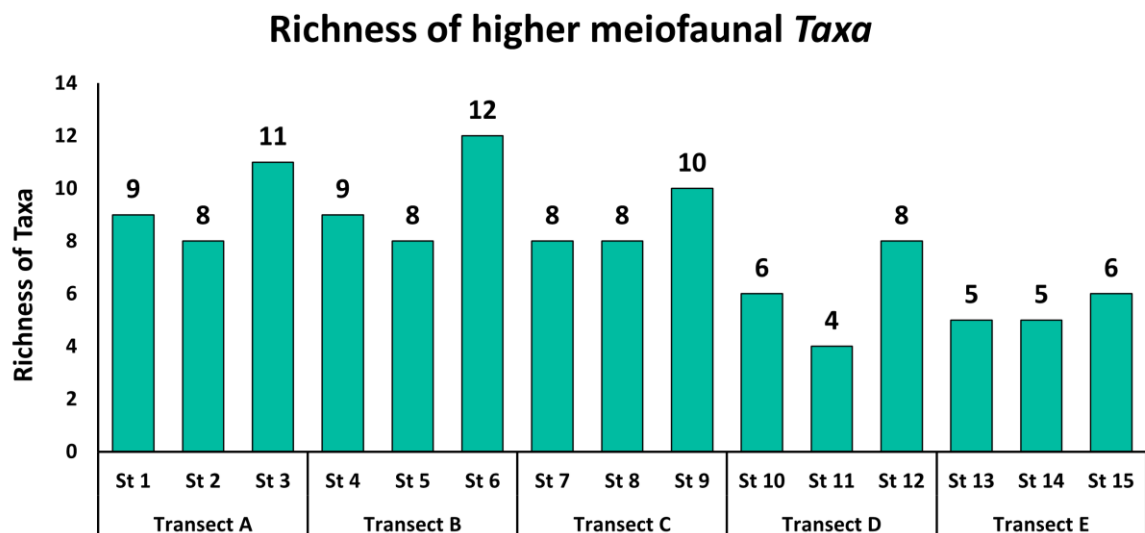


Figure 22 Meiofaunal diversity as Taxa Richness

To complement the patterns observed in *taxa* richness, a selection of representative meiofaunal organisms, isolated during sorting and photographed using a HIROX HRX-01 digital microscope, is shown in **Figure 23**.



Figure 23 Photographs taken using the HIROX HRX-01 digital microscope, in which we can observe a selection of the meiofauna taxa found within sediment samples: **A)** and **B)** Copepods; **C)** Kinorhynch; **D)** Halacaroid; **E)** Amphipod; **F)** Isopod; **G)** Tanaidacean; **H)** Cumacean.

4.3.3 Meiofaunal Assemblages Structure

Figures 24 and 25 illustrate the distribution of meiofaunal *taxa* across the stations, providing an overview of community structure.

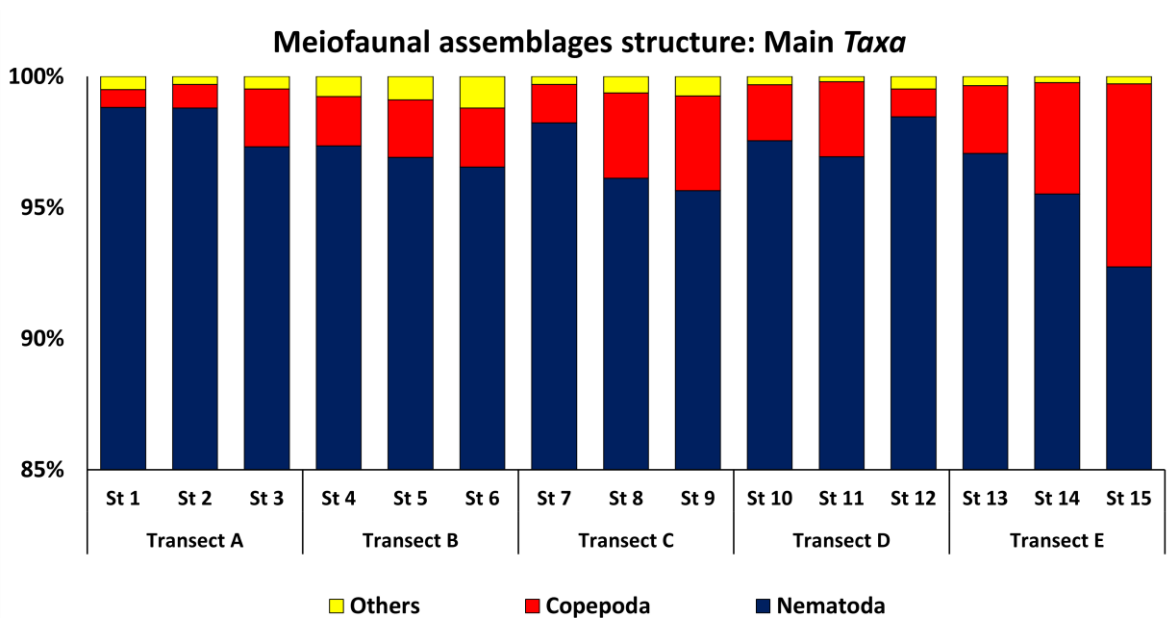


Figure 24 Meiofauna main taxa assemblages' structure

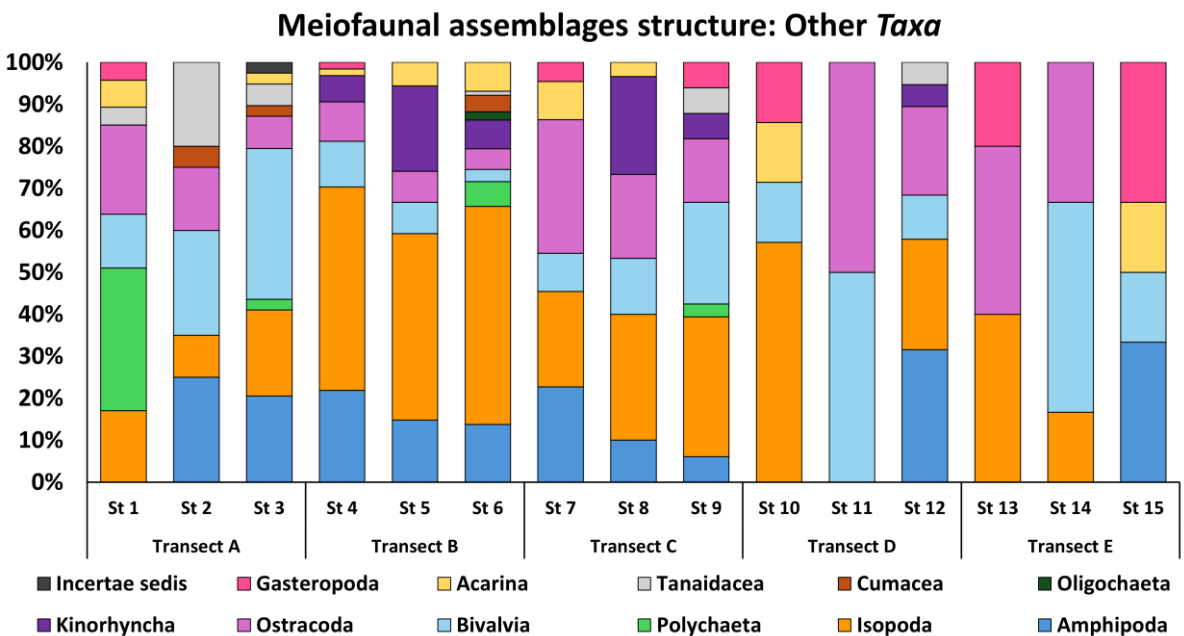


Figure 25 Meiofauna minor taxa assemblages' structure

Nematoda were the dominant *taxon* at all stations, consistently exceeding 92% of total abundance, followed by *copepoda* as the second most abundant *taxon*. All the other *taxa* contributed less than 1.2% and were classified as minor *taxa*. PERMANOVA revealed significant differences in assemblages' structure for both main and minor *taxa* ($p < 0.05$). Pairwise comparisons for the main *taxa* showed significant differences between transects A–B, A–C, A–D, A–E, B–D, and D–E ($p < 0.05$). The highest percentages of *copepoda* were observed in transect E, followed by transect C; transect B exhibited the highest proportion of minor *taxa*, whereas transect A had the highest proportion of *nematoda*. No significant differences were detected among the remaining transects ($p > 0.05$). For minor *taxa* (**Figure 25**), *isopoda* was the most abundant *taxon*, followed by *bivalvia*, *ostracoda*, and *amphipoda*. Pairwise comparisons indicated highly significant differences between transects A–B and A–C ($0.001 < p < 0.002$), and significant but less pronounced differences between transects A–D and A–E ($p = 0.015$), B–D ($p = 0.027$), and B–E ($p = 0.029$). No significant differences were detected among the remaining transects ($p > 0.05$). Notably, stations exhibiting the highest *taxa* richness (stations 6, 3, and 9) also displayed the greatest assemblage's diversity.

4.3.4 Meiofaunal Assemblages Biomass

Total meiofaunal biomass (**Figure 26**) generally reflected the patterns observed for total abundance (**Figure 21**).

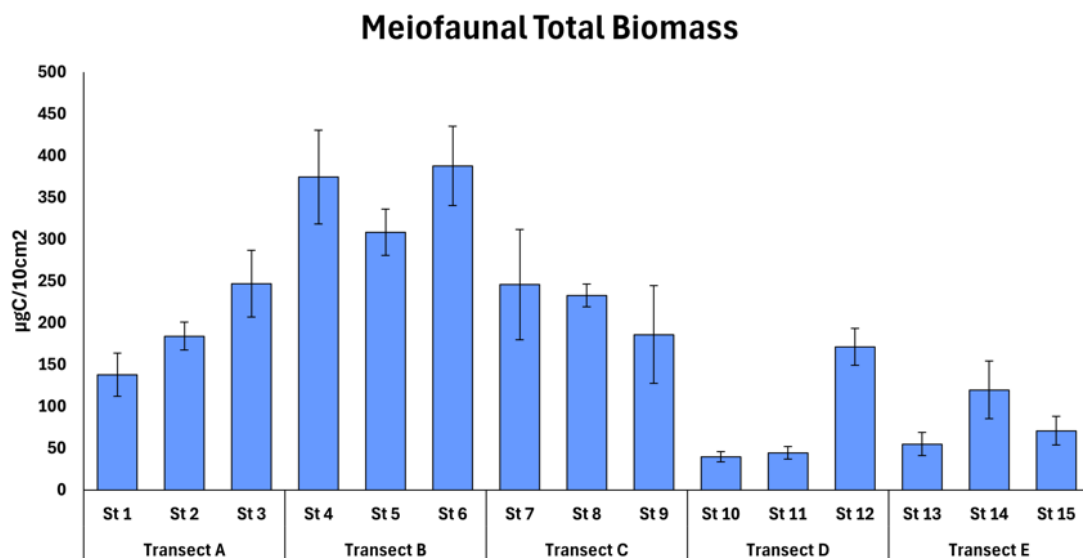


Figure 26 Meiofauna Total Biomass ($\mu\text{gC}/10\text{cm}^2$)

PERMANOVA revealed significant differences among transects ($p < 0.05$). Pairwise comparisons indicated significant differences between all transects ($p < 0.05$), except for transects A–C and D–E ($p > 0.05$). Indeed, transect B exhibited the highest biomass values, with maxima at stations 6 ($388.0 \pm 47.3 \mu\text{gC}/10 \text{ cm}^2$) and 4 ($374.6 \pm 56.2 \mu\text{gC}/10 \text{ cm}^2$), while station 5, within the same transect, showed lower values ($308.6 \pm 27.6 \mu\text{gC}/10 \text{ cm}^2$). Biomass in transects A and C was comparable, lower than in transect B but higher than in transects D and E, which recorded the lowest values, with a minimum at station 10 ($39.8 \pm 6.2 \mu\text{gC}/10 \text{ cm}^2$); at station 12 was detected a local peak of biomass ($171.4 \pm 22.0 \mu\text{gC}/10 \text{ cm}^2$).

Analysis of the contribution of *taxa* to total meiofaunal biomass (**Figure 27 and 28**) showed that *nematoda* remained the dominant *taxon*, although their contribution was lower than observed for total abundance, ranging from 47.1% at station 15 to 84.4% at station 7. *Copepoda* were the second largest contributors, accounting for 8.7% at station 12 up to 46.6% at station 5. Minor *taxa* contributed less to the total biomass than *nematoda* and *copepoda*, but their relative contribution was higher than observed for total abundance,

reaching a maximum of 16% at station 6. Among minor *taxa*, *isopoda* contributed most to biomass, followed by *ostracoda*, *bivalvia*, and *amphipoda*, consistently with patterns observed for total abundance (Figure 24).

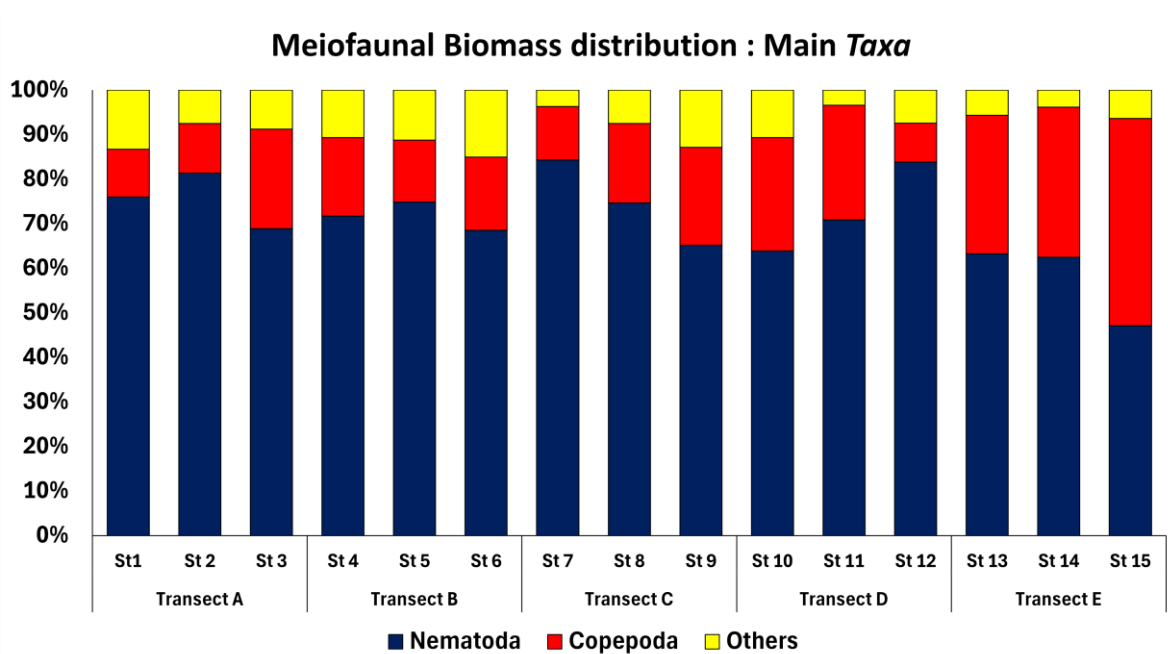


Figure 27 Meiofauna main taxa biomass distribution

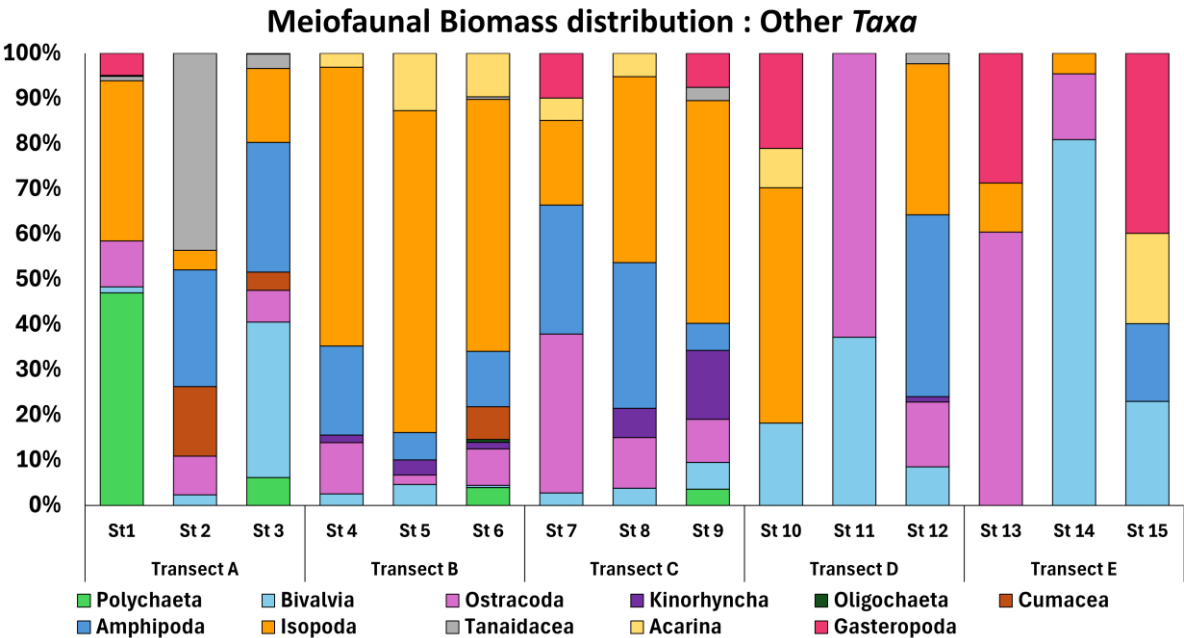


Figure 28 Meiofauna other taxa biomass distribution

4.4 Drivers for meiofaunal assemblages

DistLM analysis showed that total meiofaunal abundance was significantly influenced by the environmental variables considered (**Table 2**), with seagrass meadows having the greatest influence (55.3% of variance, $p = 0.001$), followed by depth (9.2%, $p = 0.008$), biopolymeric carbon (BPC; 5.8%, $p = 0.004$), and grain size (3.4%, $p = 0.039$). The final model explained 73.7% of the total variability in meiofaunal abundance.

Table 2 DistLM analysis results on total meiofaunal abundance. The percentage of explained variation (%Var) and cumulative explained variation (%Cum) are reported for each predictor. SS indicates sums of squares, F the pseudo-F statistic, and p the associated probability level. Reported: $p < 0.05$ (*) and $p < 0.01$ (**).

Variable	R ²	SS(trace)	Pseudo F	p-value	% Var	% Cumul
Seagrass meadow	0,55	1,70E+07	53,24	0,001**	55,3%	55,3%
Depth (m)	0,65	2,83E+06	10,93	0,008**	9,2%	64,5%
BPC (mg/g)	0,74	1,77E+06	8,77	0,004**	5,8%	70,3%
Grain Size	0,68	1,04E+06	4,31	0,039*	3,4%	73,7%

DistLM analysis showed that meiofaunal diversity as *taxa* richness was significantly influenced only by the presence of seagrass meadows, which explained 37.4% of the total variation ($p = 0.001$), with the remaining variability attributable to other factors (**Table 3**).

Table 3 DistLM analysis results on meiofaunal *taxa* richness. The percentage of explained variation (%Var) and cumulative explained variation (%Cum) are reported for each predictor. SS indicates sums of squares, F the pseudo-F statistic, and p the associated probability level. Reported: $p < 0.05$ (*) and $p < 0.01$ (**).

Variable	R ²	SS(trace)	Pseudo-F	p-value	% Var	% Cumul
Seagrass meadow	0,37	9,01E+01	25,72	0,001**	37,4%	37,4%

DistLM analysis showed that meiofaunal biomass was strongly influenced by the presence of seagrass meadows, which explained 54.7% of the total variance ($p = 0.001$), with biopolymeric carbon (BPC) contributing an additional 9.3% ($p = 0.003$), bringing the cumulative explained variance to 64.0% (**Table 4**)

Table 4 DistLM analysis results on meiofaunal total biomass. The percentage of explained variation (%Var) and cumulative explained variation (%Cum) are reported for each predictor. SS indicates sums of squares, F the pseudo-F statistic, and p the associated probability level. Reported: $p < 0.05$ (*) and $p < 0.01$ (**).

Variable	R ²	SS(trace)	Pseudo-F	p-value	% Var	% Cumul
Seagrass meadow	0,55	3,58E+05	51,85	0,001**	54,7%	54,7%
BPC (mg/g)	0,64	60604	10,79	0,003**	9,3%	64,0%

Assemblages' composition DistLM analysis distinguished main and minor *taxa*. For the main *taxa* (**Table 5**), assemblages composition was significantly correlated with several variables, above all the presence of seagrass meadows, which explained 28.7% of the variance ($p = 0.001$), followed by grain size (7.4%, $p = 0.005$), the chlorophyll-a-to-phaeopigment ratio (6.9%, $p = 0.011$), and the CPE:BPC (%) ratio (4.4%, $p = 0.038$), with the model explaining 52.6% of the total variance. For minor *taxa* (**Table 6**), seagrass meadows accounted again for the largest share of variation (23.1%, $p = 0.001$), followed by the chlorophyll-a-to-phaeopigment ratio (9.4%, $p = 0.003$) and grain size (7.0%, $p = 0.006$), with the overall model explaining 39.5% of the total variability.

Table 5 DistLM analysis results on meiofauna assemblages' composition - Main *taxa*. The percentage of explained variation (%Var) and cumulative explained variation (%Cum) are reported for each predictor. SS indicates sums of squares, F the pseudo-F statistic, and p the associated probability level. Reported: $p < 0.05$ (*) and $p < 0.01$ (**).

Variable	R ²	SS(trace)	Pseudo-F	p-value	% Var	% Cumul
Seagrass meadow	0,29	6200,9	17,32	0,001**	28,7%	28,7%
Grain Size	0,43	1600,7	5,33	0,005**	7,4%	36,1%
Chl-a/Phaeo	0,36	1495,6	4,52	0,011*	6,9%	43,0%
CPE/BPC%	0,52	950,1	3,36	0,038*	4,4%	52,6%

Table 6 DistLM analysis results on meiofauna assemblages' composition - Minor taxa. The percentage of explained variation (%Var) and cumulative explained variation (%Cum) are reported for each predictor. SS indicates sums of squares, F the pseudo-F statistic, and p the associated probability level. Reported: $p < 0.05$ (*) and $p < 0.01$ (**).

Variable	R ²	SS(trace)	Pseudo-F	p-value	% Var	% Cumul
Seagrass meadow	0,23	9901,7	12,93	0,001**	23,1%	23,1%
Chl-a/Phaeo	0,32	4008,9	5,82	0,003**	9,4%	32,5%
Grain Size	0,39	2989,8	4,73	0,006**	7,0%	39,5%

The NMDS ordination based on meiofaunal abundance (**Figure 29**) revealed a separation between samples from vegetated and unvegetated sites, which clustered distinctly, indicating marked differences in assemblages structure associated with the presence of seagrass meadows (stress = 0.177).

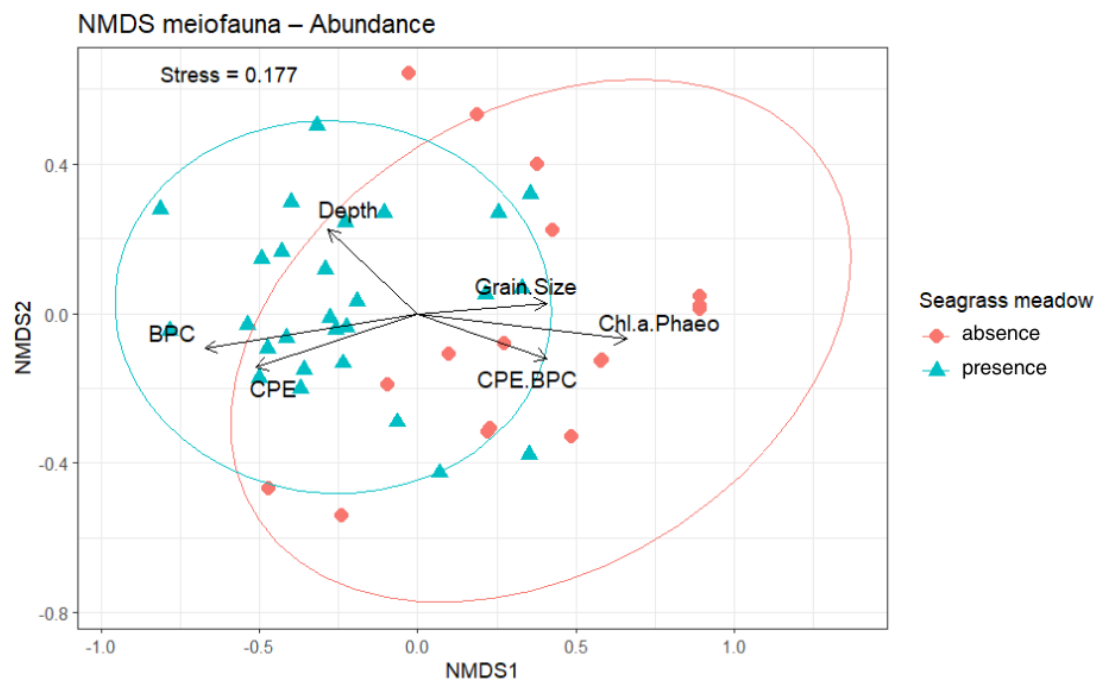


Figure 29 Non-metric multidimensional scaling (NMDS) ordination based on Bray–Curtis dissimilarities of meiofauna abundance, with environmental variables fitted as vectors.

Environmental vectors indicated that meiofaunal abundance patterns were primarily associated with seagrass presence, followed by depth, sediment trophic variables and grain size.

The NMDS ordination based on meiofaunal biomass (**Figure 30**) also showed a separation between samples from vegetated and unvegetated sediments, with them clustering distinctly (stress = 0.205), confirming that seagrass presence significantly influences biomass distribution.

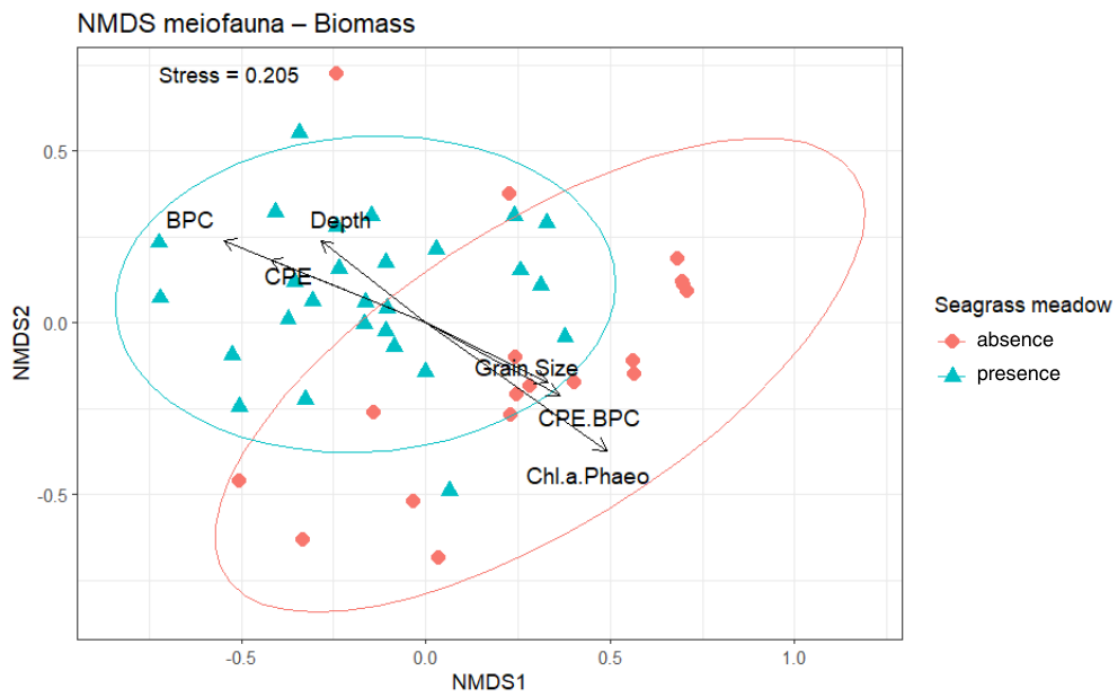


Figure 30 Non-metric multidimensional scaling (NMDS) ordination based on Bray–Curtis dissimilarities of meiofauna biomass, with environmental variables fitted as vectors.

The projection of environmental vectors revealed that biomass patterns were closely related to seagrass presence above all, but also to sediment trophic variables, while grain size and depth had lower influence.

5. DISCUSSION

5.1 Meiofaunal distribution: a comparison between vegetated and unvegetated habitats

The results of this study showed clear differences in both environmental variables and meiofaunal descriptors, between vegetated and unvegetated sediments. The study area displayed, in fact, higher silt-clay fraction ($< 63 \mu\text{m}$) and higher grain size heterogeneity in vegetated sediments than in unvegetated sediments. Moreover, higher biopolymeric carbon (BPC) and total phytopigment concentrations (given by the sum of chlorophyll-a and phaeopigment concentrations) were recorded in vegetated sediments compared with unvegetated ones, indicating a substantial accumulation of primary biomass, linked to the presence of seagrass meadows.

Seagrasses, as ecosystem engineers, modify the habitat increasing its complexity and giving it a three-dimensional structure (Unsworth et al., 2019). They also modify the chemical, physical and biological characteristics of the surrounding environment, such as the grain size composition of the sediment and the availability and distribution of the organic material (Thathsarani et al., 2025; Unsworth et al., 2026). Indeed, sediments hosting seagrass meadows generally accumulate a greater amount of finer sediment particles. This finer particles of sediment tend to accumulate more organic carbon, favouring the aggregation and deposition of organic matter in vegetated sediments (Dahl et al., 2016). This highlights how seagrasses influence the grain size composition and the organic matter distribution of the area they cover.

Comparison with previous studies conducted in the Gabicce Mare shallow coastal area revealed similar results for both grain size composition and organic matter distribution. As regards grain size composition, both in the current study and in previous ones, fine and

very fine sand fractions dominated the sediment composition, with a higher fraction of silt-clay in vegetated sediments (DISVA, 2011; Carugati, 2018; Università degli Studi di Ancona, 2002). Similar patterns in total phytopigments and biopolymeric carbon (BPC) concentrations were also found among the different studies with generally higher concentrations in vegetated sediments rather than unvegetated ones (Università degli Studi di Ancona, 2002; Da Ros et al., 2021). In particular, the highest BPC and total phytopigments accumulations, found in this present study along the fully vegetated transect B and C, agreed with the previous investigations conducted in Gabicce Mare (Università degli Studi di Ancona, 2002; Carugati, 2019). The trophic quality indicators of organic matter did not show significant differences between vegetated and unvegetated sediments. However, the positive values of the protein-to-carbohydrate ratio across all transects indicated an overall high quality and availability of the organic matter within the investigated area. In fact, proteins, which dominate carbohydrates in the whole study area, are mobilized more quickly than the latter, being more rapidly available to benthic consumers (Dell'Anno et al., 2002). Furthermore, no significant differences related to the bathymetric gradient between vegetated and unvegetated sediments were detected in any of the environmental variable's analyses. This may be partly attributed to the limited bathymetric range along the transects within the study area, with a maximum depth-difference of 2.1 m.

The results of the present study indicate that meiofaunal abundance, biomass, diversity and assemblage composition, along the whole study area, were influenced by the presence or absence of seagrass meadows. In fact, higher values of total abundance, richness of higher meiofaunal *taxa* and biomass were detected along transects being entirely within the seagrass meadows boundaries (transects B and C), at station 2 and 3 of transect A and at station 12 of transect D. Station 12 was particularly significant for this study, representing the only

vegetated station along transect D. It showed marked increases in all meiofaunal descriptors values compared with the adjacent unvegetated stations of the same transect. These results indicated that vegetated sediments supported higher meiofaunal abundance, diversity as *taxa* richness and biomass, likely as a consequence of the increased substrate complexity provided by seagrass leaves, roots and rhizomes, which enhance sediment stabilization, organic matter retention and refuge availability (Orth et al., 2006; Cullen-Unsworth & Unsworth, 2013). Indeed, sediment grain size composition and biopolymeric carbon distribution are influenced by the presence or absence of seagrasses and can, in turn, influence the distribution of meiofaunal assemblages. Higher grain size heterogeneity is generally associated with higher biodiversity, in agreement with the demonstrated relationship between sediment texture, interstitial space and meiofaunal distribution (Coull, 1988; Giere, 2009). Grain size heterogeneity can increase microhabitat availability (Gingold et al., 2010), thus enhancing meiofaunal diversity. Similarly, several studies demonstrated that biopolymeric carbon distribution, and consequently the quantity and availability of potential food, play an important role in shaping benthic assemblages (Albertelli et al., 1999; Bianchelli et al., 2024; Giere, 2009; Pusceddu et al., 2009).

Assemblages composition analysis showed the strong dominance of *nematoda* and secondarily *copepoda*, across all transects, reflecting their ecological versatility and central role within benthic food webs (Coull, 1988; Giere, 2009). Minor *taxa* displayed a more heterogeneous spatial distribution along transects, suggesting greater sensitivity to habitat structure and local environmental conditions (Schratzberger & Ingels, 2018), with the most diverse assemblages structures recorded in vegetated sediments. In this study, moreover, different *taxa* were found to be more closely associated with vegetated sediments while others with unvegetated sediments. *Kinorhyncha* and *cumacea* were found exclusively

associated with vegetated sediments, while *taxa* such as *isopoda*, *amphipoda*, *acarina* and *tanaidacea* occurred in both sediment types but reached higher abundance and biomass contribution in presence of seagrasses. Conversely, *gastropoda* and *bivalvia* showed higher abundance and biomass contribution in unvegetated sediments. These patterns were coherent with literature describing *taxon*-specific habitat affinities in meiofaunal assemblages (Giere, 2009).

Higher meiofaunal abundance, biomass and diversity within seagrass meadows, compared with nearby unvegetated sediments, have been reported in several studies performed in different parts of the world (Baldrighi et al., 2021; Coccozza di Montanara et al., 2024; Fonseca et al., 2011; Liu et al., 2025; Pusceddu et al., 2016) but also within this same study area, showing the highest meiofaunal abundance, biomass and diversity along areas corresponding to transect B and C of the present investigation, matching with the present study results (DISVA, 2011; Da Ros et al., 2021; Carugati et al., 2018). The total number of meiofaunal *taxa* found in the Gabicce Mare area was similar in the various studies, as well as meiofaunal main *taxa* assemblages' composition, with strong dominance of *nematoda* and then of *copepoda*, while minor *taxa* exhibited greater variability in their composition across studies. This can be explained with potential differences in environmental conditions, temporal variability and sampling approaches between the compared studies (DISVA, 2011; Da Ros et al., 2021; Carugati et al., 2018).

5.2 Drivers of meiofaunal assemblages

DistLM analyses were performed to explain which of the various environmental variables and to what extent they explain meiofauna variations in terms of abundance, diversity, biomass and assemblages' composition, inside the study area. In all the cases, the presence or absence of seagrass meadows has been recognised as the main factor influencing meiofaunal assemblages. This was also supported by NMDS ordinations, showing the segregation of samples between vegetated and unvegetated sediments. Among the other investigated environmental variables, however, both sediment grain size and trophic availability were found to influence meiofaunal assemblages, but to a lesser extent than the presence or absence of seagrasses. Depth, by contrast, was found to influence meiofaunal assemblages only in terms of total abundance.

Based on this evidence, the presence or absence of seagrass meadows appeared to be the main driver of meiofaunal biodiversity within the analysed area, while sediment grain size and trophic availability appeared to act as secondary drivers, as they are also influenced by the presence or absence of seagrasses. Seagrass meadows increase habitat heterogeneity and niche availability, allowing a higher number of *taxa* to coexist in vegetated sediments compared with unvegetated ones (Baldrighi et al., 2021; Ferretto et al., 2023), while trophic and grain size variables of the sediment, which are themselves influenced by seagrasses, contribute to influence meiofaunal assemblages mainly in terms of abundance and biomass (Bianchelli et al., 2024).

Overall, the results of this study highlight the close relationship between meiofaunal assemblages and seagrass meadows, underlining the ecological importance of these habitats in sustaining benthic biodiversity and ecosystem functioning. Meiofauna plays a central role

in organic matter processing, nutrient regeneration and energy transfer to higher trophic levels (Coull, 1988; Danovaro et al., 2007). Consequently, the decline of seagrass meadows may trigger cascading effects on coastal food webs, compromising various ecosystem services (do Amaral Camara Lima et al., 2023).

Given the current rate of global decline of seagrass meadows (de Los Santos et al., 2019; Waycott et al., 2009), their protection, conservation and restoration should be considered a priority. Nevertheless, despite being formally recognized as endangered ecosystems, seagrass meadows are often not perceived as “iconic habitats” (unlike coral reefs, mangroves, etc.), resulting in lower public awareness and fewer resources dedicated to their conservation and restoration (Da Ros et al., 2021)

6. CONCLUSIONS

This study highlighted the pivotal role of seagrass meadows in structuring coastal benthic ecosystems, demonstrating how habitat complexity, sediment grain size and trophic conditions are interconnected and act together on meiofaunal assemblages.

The analyses and interpretation of meiofaunal assemblages' descriptors and environmental variables, together with multivariate statistical analyses, led to the conclusion that, within the Gabicce Mare shallow coastal area, the main driver shaping meiofaunal assemblages is the presence or absence of seagrass meadows.

There are also secondary drivers, specifically sediment grain size composition and trophic availability, which are themselves influenced by the presence or absence of seagrasses, having important influence on meiofaunal assemblages. In fact, higher grain size heterogeneity and presence of finer sediment particles, were found in vegetated sediments as well as higher concentrations of organic matter, both considering biopolymeric carbon and phytopigments.

The trophic quality of the organic matter was found to be high throughout the study area, without showing significant differences between vegetated and unvegetated sediments.

Meiofaunal abundance, biomass and diversity as *taxa* richness were significantly higher in vegetated sediments, confirming seagrass meadows as habitat-forming species, enhancing environmental heterogeneity and niche availability. Moreover, assemblages structure of meiofaunal assemblages showed, in all the transects, *nematoda* as the very dominant *taxon*, always followed by *copepoda*, emphasising their ecological versatility, resilience to environmental changes and central role in benthic food webs.

All these results, combined with the other data collected as part of the DiverSea project (macrofauna, molecular, e-DNA, microbiological, visual and photo census data, etc.) can contribute to a more comprehensive understanding of seagrass coastal systems ecology. They could also provide valuable insights for the conservation and management of these fundamental ecosystems, whose loss entails not only a reduction in biodiversity but also a huge loss of ecosystem goods and services, with repercussions on economy as well, underlining the urgency of the situation.

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